Supplementary Appendix 3

NCPE

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1 Importance sampling: model diagnostics, parameter distributions, and survival estimates

The tables and figures in this section show survival curve parameter estimates, covariance structures, importance sampling diagnostics, and (probabilistic) survival curve predictions, obtained from trial data only estimated using maximum likelihood estimation (MLE), and combining trial data with external information using the importance sampling algorithm described in the methods section of the main article and in Appendix 1.

1.1 Comparisons of parameter estimates with and without external information

The below tables compare parameter estimates and covariance matrices obtained from trial data only with those obtained combining trial data with external information, for each of the survival models under consideration.

Table 1: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	x		x
rate	-4.562874	rate	-4.467088

Table 2: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	rate		rate
rate	0.0169492	rate	0.0138444

Table 3: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	x		x
shape	0.2028128	shape	0.2199019
scale	4.2127617	scale	4.1750757

Table 4: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	shape	scale		shape	scale
shape	0.0147844	-0.0228526	shape	0.0094519	-0.0118402
scale	-0.0228526	0.0466214	scale	-0.0118402	0.0235770

Table 5: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	x		x
shape	0.0389625	shape	0.0197027
rate	-4.8682858	rate	-4.7220930

Table 6: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	shape	rate		shape	rate
shape	0.0006747	-0.0056127	shape	0.0000970	-0.0013281
rate	-0.0056127	0.0636371	rate	-0.0013281	0.0316261

Table 7: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	x		x
meanlog	4.5379239	meanlog	4.2299230
sdlog	0.5475716	sdlog	0.4404943

Table 8: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	meanlog	sdlog		meanlog	sdlog
meanlog	0.0696811	0.0247171	meanlog	0.0545435	0.0201378
sdlog	0.0247171	0.0114622	sdlog	0.0201378	0.0100655

Table 9: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	x		x
shape	0.2450486	shape	0.3103378
scale	4.0866536	scale	3.9558418

Table 10: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	shape	scale
shape	0.0144929	-0.0213003
scale	-0.0213003	0.0434085

	shape	scale
shape	0.0111243	-0.0139387
scale	-0.0139387	0.0276612

Table 11: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

	x
mu	4.1703317
sigma	-0.3599072
Q	1.2031485

	x
mu	4.1857610
sigma	-0.3818327
Q	1.2508022

1.2 Effective sample size plots

The below plots show the estimated effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the survival distributions under consideration. The vertical line shows the optimal value of alpha identified by the algorithm. Larger ESS values are better.

$data alpha ESS 1 0.5000000 1374.742 2 0.5477387 1721.881 3 0.5954774 1451.175 4 0.6432161 1674.855 5 0.6909548 2106.190 6 0.7386935 2523.770 7 0.7864322 2649.020 8 0.8341709 2762.528 9 0.8819095 2905.483 10 0.9296482 3046.724 11 0.9773869 3166.361 12 1.0251256 3286.473 13 1.0728643 3326.243 14 1.1206030 3377.873 15 1.1683417 3434.576 16 1.2160804 3422.563 17 1.2638191 3469.566 18 1.3115578 3482.941 19 1.3592965 3509.570 20 1.4070352 3536.320 21 1.4547739 3537.769 22 1.5025126 3552.132 23 1.5502513 3537.335 24 1.5979899 3540.887 25 1.6457286 3542.386 26 1.6934673 3515.282 27 1.7412060 3474.743 28 1.7889447 3497.096 29 1.8366834 3461.971 30 1.8844221 3462.069 31 1.9321608 3436.331 32 1.9798995 3453.571 33 2.0276382 3462.713 34 2.0753769 3401.592 35 2.1231156 3398.429 36 2.1708543 3386.580 37 2.2185930 3338.533 38 2.2663317 3369.317 39 2.3140704 3325.581 40 2.3618090 3326.783 41 2.4095477 3295.373 42 2.4572864 3322.869 43 2.5050251 3265.954 44 2.5527638 3267.504 45 2.6005025 3215.345 46 2.6482412 3232.613 47 2.6959799 3178.101 48 2.7437186 3199.021 49 2.7914573 3228.472 50 2.8391960 3157.732 51 2.8869347 3142.424 52 2.9346734 3116.167 53 2.9824121 3117.909 54 3.0301508 3100.144 55 3.0778894 3128.914 56 3.1256281 3118.109 57 3.1733668 3106.083 58 3.2211055 3007.866 59 3.2688442 2974.351 60 3.3165829 3002.724 61 3.3643216 3010.000 62 3.4120603 2960.332 63 3.4597990 2946.748 64 3.5075377 2940.971 65 3.5552764 2948.102 66 3.6030151 2923.501 67 3.6507538 2854.046 68 3.6984925 2900.584 69 3.7462312 2881.166 70 3.7939698 2832.472 71 3.8417085 2858.870 72 3.8894472 2824.555 73 3.9371859 2820.594 74 3.9849246 2874.733 75 4.0326633 2779.490 76 4.0804020 2758.824 77 4.1281407 2739.248 78 4.1758794 2768.897 79 4.2236181 2751.999 80 4.2713568 2773.809 81 4.3190955 2745.863 82 4.3668342 2677.415 83 4.4145729 2691.683 84 4.4623116 2670.363 85 4.5100503 2735.073 86 4.5577889 2672.560 87 4.6055276 2661.627 88 4.6532663 2623.070 89 4.7010050 2601.185 90 4.7487437 2641.303 91 4.7964824 2638.838 92 4.8442211 2627.652 93 4.8919598 2588.146 94 4.9396985 2595.313 95 4.9874372 2593.254 96 5.0351759 2621.082 97 5.0829146 2523.287 98 5.1306533 2514.120 99 5.1783920 2550.270 100 5.2261307 2516.684 101 5.2738693 2526.364 102 5.3216080 2539.590 103 5.3693467 2470.404 104 5.4170854 2487.488 105 5.4648241 2524.850 106 5.5125628 2511.322 107 5.5603015 2534.604 108 5.6080402 2469.232 109 5.6557789 2486.036 110 5.7035176 2485.887 111 5.7512563 2452.091 112 5.7989950 2439.583 113 5.8467337 2380.998 114 5.8944724 2395.854 115 5.9422111 2397.151 116 5.9899497 2380.993 117 6.0376884 2405.771 118 6.0854271 2378.839 119 6.1331658 2390.620 120 6.1809045 2437.273 121 6.2286432 2301.790 122 6.2763819 2343.476 123 6.3241206 2352.255

Table 12: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

	mu	sigma	Q
mu	0.1026807	0.2899736	-0.3589428
sigma	0.2899736	1.2303463	-1.6290913
Q	-0.3589428	-1.6290913	2.1837103

	mu	sigma	Q
mu	0.0219135	0.0031467	0.0125013
sigma	0.0031467	0.2238160	-0.3291366
Q	0.0125013	-0.3291366	0.5057176

124 6.3718593 2326.194 125 6.4195980 2370.657 126 6.4673367 2397.246 127 6.5150754 2360.861 128 6.5628141 2298.323 129 6.6105528 2254.071 130 6.6582915 2290.810 131 6.7060302 2275.889 132 6.7537688 2276.802 133 6.8015075 2301.826 134 6.8492462 2234.672 135 6.8969849 2315.448 136 6.9447236 2255.052 137 6.9924623 2267.353 138 7.0402010 2256.876 139 7.0879397 2196.563 140 7.1356784 2205.132 141 7.1834171 2254.306 142 7.2311558 2218.106 143 7.2788945 2229.722 144 7.3266332 2207.257 145 7.3743719 2262.649 146 7.4221106 2201.430 147 7.4698492 2145.495 148 7.5175879 2231.051 149 7.5653266 2168.183 150 7.6130653 2151.755 151 7.6608040 2171.443 152 7.7085427 2154.998 153 7.7562814 2154.303 154 7.8040201 2172.973 155 7.8517588 2168.450 156 7.8994975 2172.803 157 7.9472362 2066.050 158 7.9949749 2076.939 159 8.0427136 2166.771 160 8.0904523 2105.380 161 8.1381910 2147.535 162 8.1859296 2131.798 163 8.2336683 2133.578 164 8.2814070 2092.964 165 8.3291457 2101.405 166 8.3768844 2108.381 167 8.4246231 2103.686 168 8.4723618 2131.087 169 8.5201005 2079.216 170 8.5678392 2090.796 171 8.6155779 2071.576 172 8.6633166 2043.111 173 8.7110553 2024.022 174 8.7587940 2022.501 175 8.8065327 1976.447 176 8.8542714 2054.155 177 8.9020101 2039.249 178 8.9497487 2021.136 179 8.9974874 2032.214 180 9.0452261 1998.072 181 9.0929648 1994.595 182 9.1407035 2056.321 183 9.1884422 1989.607 184 9.2361809 1992.745 185 9.2839196 1996.265 186 9.3316583 1963.904 187 9.3793970 1963.061 188 9.4271357 1971.407 189 9.4748744 1975.000 190 9.5226131 1979.277 191 9.5703518 1978.310 192 9.6180905 1895.980 193 9.6658291 1997.564 194 9.7135678 1930.372 195 9.7613065 1958.295 196 9.8090452 1927.530 197 9.8567839 1977.222 198 9.9045226 1928.242 199 9.9522613 1916.788 200 10.0000000 1950.167

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*labels*xintercept [1] “xintercept”

attr(,“class”) [1] “gg” “ggplot”

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*layers*layers[[1]] geom_line: na.rm = FALSE, orientation = NA stat_identity: na.rm = FALSE position_identity

$layers[[2]] mapping: xintercept = ~xintercept geom_vline: na.rm = FALSE stat_identity: na.rm = FALSE position_identity

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*mapping*x expr: ^alpha env: 0x000001f377f59820

$y expr: ^ESS env: 0x000001f377f59820

attr(,"class") [1] "uneval"

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*labels*xintercept [1] "xintercept"

attr(,"class") [1] "gg" "ggplot"

$data alpha ESS 1 0.5000000 1091.5160 2 0.5477387 1369.4997 3 0.5954774 1398.8758 4 0.6432161 1545.6417 5 0.6909548 1638.6281 6 0.7386935 1670.3824 7 0.7864322 1672.9770 8 0.8341709 1705.1042 9 0.8819095 1714.6201 10 0.9296482 1715.6126 11 0.9773869 1704.8392 12 1.0251256 1686.4979 13 1.0728643 1711.4092 14 1.1206030 1674.2041 15 1.1683417 1655.8591 16 1.2160804 1664.0720 17 1.2638191 1564.7503 18 1.3115578 1487.3723 19 1.3592965 1483.4641 20 1.4070352 1499.0867 21 1.4547739 1496.3985 22 1.5025126 1406.9376 23 1.5502513 1427.3707 24 1.5979899 1408.2070 25 1.6457286 1400.3112 26 1.6934673 1347.5779 27 1.7412060 1385.1032 28 1.7889447 1296.1976 29 1.8366834 1322.9622 30 1.8844221 1249.3825 31 1.9321608 1287.6859 32 1.9798995 1236.9198 33 2.0276382 1197.1253 34 2.0753769 1222.2967 35 2.1231156 1152.3907 36 2.1708543 1129.6656 37 2.2185930 1155.8954 38 2.2663317 1109.3164 39 2.3140704 1102.9577 40 2.3618090 1079.2594 41 2.4095477 1113.1697 42 2.4572864 1023.8170 43 2.5050251 1025.8709 44 2.5527638 1040.8588 45 2.6005025 978.9620 46 2.6482412 1012.1871 47 2.6959799 983.9128 48 2.7437186 984.6743 49 2.7914573 958.5666 50 2.8391960 938.6626 51 2.8869347 936.0246 52 2.9346734 915.9185 53 2.9824121 928.5741 54 3.0301508 906.8314 55 3.0778894 933.5697 56 3.1256281 868.1744 57 3.1733668 864.1574 58 3.2211055 859.8028 59 3.2688442 868.4061 60 3.3165829 868.5983 61 3.3643216 829.1071 62 3.4120603 829.8111 63 3.4597990 821.2504 64 3.5075377 792.8195 65 3.5552764 809.9781 66 3.6030151 769.8534 67 3.6507538 748.4211 68 3.6984925 735.9870 69 3.7462312 728.7494 70 3.7939698 734.2594 71 3.8417085 719.8643 72 3.8894472 743.0309 73 3.9371859 723.2655 74 3.9849246 715.9102 75 4.0326633 741.8497 76 4.0804020 702.4233 77 4.1281407 680.7756 78 4.1758794 657.9461 79 4.2236181 685.6244 80 4.2713568 676.9753 81 4.3190955 662.8679 82 4.3668342 674.3829 83 4.4145729 624.1147 84 4.4623116 691.7307 85 4.5100503 644.3327 86 4.5577889 628.4081 87 4.6055276 670.8163 88 4.6532663 638.2374 89 4.7010050 606.0939 90 4.7487437 623.8207 91 4.7964824 598.1556 92 4.8442211 610.8705 93 4.8919598 631.4864 94 4.9396985 629.0799 95 4.9874372 623.8980 96 5.0351759 582.2468 97 5.0829146 603.2929 98 5.1306533 584.7559 99 5.1783920 564.4161 100 5.2261307 567.9921 101 5.2738693 540.7903 102 5.3216080 556.4421

103 5.3693467 558.4789 104 5.4170854 571.8405 105 5.4648241 510.4472 106 5.5125628 557.6788 107 5.5603015 536.1733 108 5.6080402 568.0376 109 5.6557789 515.9147 110 5.7035176 530.8192 111 5.7512563 543.9747 112 5.7989950 542.7290 113 5.8467337 504.2292 114 5.8944724 505.9214 115 5.9422111 522.1367 116 5.9899497 494.2037 117 6.0376884 484.9512 118 6.0854271 508.5068 119 6.1331658 480.0842 120 6.1809045 493.3658 121 6.2286432 471.4323 122 6.2763819 481.6313 123 6.3241206 494.4031 124 6.3718593 485.0870 125 6.4195980 433.4563 126 6.4673367 458.0690 127 6.5150754 467.2765 128 6.5628141 462.7114 129 6.6105528 475.2591 130 6.6582915 464.7613 131 6.7060302 452.3605 132 6.7537688 436.2601 133 6.8015075 451.7548 134 6.8492462 431.9157 135 6.8969849 437.5417 136 6.9447236 458.1287 137 6.9924623 431.8605 138 7.0402010 444.2302 139 7.0879397 445.2560 140 7.1356784 425.9852 141 7.1834171 443.0908 142 7.2311558 425.2051 143 7.2788945 408.0466 144 7.3266332 406.0754 145 7.3743719 436.6157 146 7.4221106 415.7501 147 7.4698492 413.9541 148 7.5175879 414.8614 149 7.5653266 403.0377 150 7.6130653 408.3175 151 7.6608040 390.5283 152 7.7085427 372.4082 153 7.7562814 404.4487 154 7.8040201 387.4469 155 7.8517588 369.7017 156 7.8994975 383.3292 157 7.9472362 394.9250 158 7.9949749 370.9000 159 8.0427136 369.7130 160 8.0904523 380.5199 161 8.1381910 361.0940 162 8.1859296 379.9333 163 8.2336683 355.1172 164 8.2814070 397.0520 165 8.3291457 342.3992 166 8.3768844 366.4622 167 8.4246231 397.3490 168 8.4723618 371.5997 169 8.5201005 368.8471 170 8.5678392 334.8268 171 8.6155779 367.0269 172 8.6633166 360.4508 173 8.7110553 355.1155 174 8.7587940 370.0379 175 8.8065327 358.3617 176 8.8542714 361.1302 177 8.9020101 344.9037 178 8.9497487 351.4557 179 8.9974874 331.0129 180 9.0452261 347.0565 181 9.0929648 347.8150 182 9.1407035 345.9979 183 9.1884422 346.3481 184 9.2361809 331.4696 185 9.2839196 326.3378 186 9.3316583 302.5592 187 9.3793970 349.6199 188 9.4271357 351.8922 189 9.4748744 340.9014 190 9.5226131 331.9592 191 9.5703518 330.5660 192 9.6180905 313.2577 193 9.6658291 287.7517 194 9.7135678 343.8003 195 9.7613065 335.2036 196 9.8090452 317.8904 197 9.8567839 304.0329 198 9.9045226 330.6175 199 9.9522613 293.3395 200 10.0000000 305.7318

*layers*layers[[1]] geom_line: na.rm = FALSE, orientation = NA stat_identity: na.rm = FALSE position_identity

$layers[[2]] mapping: xintercept = ~xintercept geom_vline: na.rm = FALSE stat_identity: na.rm = FALSE position_identity

$scales <ggproto object: Class ScalesList, gg> add: function add_defaults: function add_missing: function backtransform_df: function clone: function find: function get_scales: function has_scale: function input: function map_df: function n: function non_position_scales: function scales: list set_palettes: function train_df: function transform_df: function super: <ggproto object: Class ScalesList, gg>

$guides <Guides[0] ggproto object>

*mapping*x expr: ^alpha env: 0x000001f37731c078

$y expr: ^ESS env: 0x000001f37731c078

attr(,"class") [1] "uneval"

$theme list()

$coordinates <ggproto object: Class CoordCartesian, Coord, gg> aspect: function backtransform_range: function clip: on default: TRUE distance: function draw_panel: function expand: TRUE is_free: function is_linear: function labels: function limits: list modify_scales: function range: function render_axis_h: function render_axis_v: function render_bg: function render_fg: function reverse: none setup_data: function setup_layout: function setup_panel_guides: function setup_panel_params: function setup_params: function train_panel_guides: function transform: function super: <ggproto object: Class CoordCartesian, Coord, gg>

$facet <ggproto object: Class FacetNull, Facet, gg> attach_axes: function attach_strips: function compute_layout: function draw_back: function draw_front: function draw_labels: function draw_panel_content: function draw_panels: function finish_data: function format_strip_labels: function init_gtable: function init_scales: function map_data: function params: list set_panel_size: function setup_data: function setup_panel_params: function setup_params: function shrink: TRUE train_scales: function vars: function super: <ggproto object: Class FacetNull, Facet, gg>

$plot_env <environment: 0x000001f37731c078>

$layout <ggproto object: Class Layout, gg> coord: NULL coord_params: list facet: NULL facet_params: list finish_data: function get_scales: function layout: NULL map_position: function panel_params: NULL panel_scales_x: NULL panel_scales_y: NULL render: function render_labels: function reset_scales: function

resolve_label: function setup: function setup_panel_guides: function setup_panel_params: function train_position: function super: <ggproto object: Class Layout, gg>

*labels*labels$title [1] “Effective sample size - Gompertz”

*labels*x [1] “alpha”

*labels*y [1] “ESS”

*labels*xintercept [1] “xintercept”

attr(,“class”) [1] “gg” “ggplot”

$data alpha ESS 1 0.5000000 137.9742 2 0.5477387 454.6420 3 0.5954774 360.9980 4 0.6432161 745.1240 5 0.6909548 539.0069 6 0.7386935 856.5942 7 0.7864322 906.7221 8 0.8341709 1092.2207 9 0.8819095 1285.8945 10 0.9296482 1431.9933 11 0.9773869 1488.6307 12 1.0251256 1607.2031 13 1.0728643 1770.3100 14 1.1206030 1799.2189 15 1.1683417 1918.3617 16 1.2160804 1953.1908 17 1.2638191 2052.0783 18 1.3115578 2059.5493 19 1.3592965 2085.1815 20 1.4070352 2091.3283 21 1.4547739 2175.8700 22 1.5025126 2159.8097 23 1.5502513 2216.2298 24 1.5979899 2200.8502 25 1.6457286 2194.2150 26 1.6934673 2210.5299 27 1.7412060 2224.3945 28 1.7889447 2206.9645 29 1.8366834 2207.4823 30 1.8844221 2264.3129 31 1.9321608 2218.2670 32 1.9798995 2225.4196 33 2.0276382 2158.9782 34 2.0753769 2138.0145 35 2.1231156 2197.9996 36 2.1708543 2180.9332 37 2.2185930 2100.7152 38 2.2663317 2105.5467 39 2.3140704 2069.3118 40 2.3618090 2140.2589 41 2.4095477 2110.9661 42 2.4572864 2100.7034 43 2.5050251 2000.5596 44 2.5527638 2009.8785 45 2.6005025 2010.2260 46 2.6482412 2000.2116 47 2.6959799 2015.1406 48 2.7437186 1987.8787 49 2.7914573 1946.0291 50 2.8391960 1953.3294 51 2.8869347 1945.3348 52 2.9346734 1840.7118 53 2.9824121 1921.5973 54 3.0301508 1917.1051 55 3.0778894 1878.6596 56 3.1256281 1859.2873 57 3.1733668 1840.7949 58 3.2211055 1830.3875 59 3.2688442 1756.3224 60 3.3165829 1759.2424 61 3.3643216 1801.4865 62 3.4120603 1749.2102 63 3.4597990 1749.6431 64 3.5075377 1724.1060 65 3.5552764 1748.7306 66 3.6030151 1723.4948 67 3.6507538 1657.7293 68 3.6984925 1715.4098 69 3.7462312 1647.7229 70 3.7939698 1634.8378 71 3.8417085 1663.5390 72 3.8894472 1613.7004 73 3.9371859 1592.0764 74 3.9849246 1600.0594 75 4.0326633 1580.1747 76 4.0804020 1609.7539 77 4.1281407 1597.8003 78 4.1758794 1556.3434 79 4.2236181 1593.5948 80 4.2713568 1552.0286 81 4.3190955 1521.4307 82 4.3668342 1494.7847 83 4.4145729 1460.6346 84 4.4623116 1489.2424 85 4.5100503 1471.4323 86 4.5577889 1494.3926 87 4.6055276 1453.3690 88 4.6532663 1503.2633 89 4.7010050 1420.7329 90 4.7487437 1394.2841 91 4.7964824 1440.7461 92 4.8442211 1403.1798 93 4.8919598 1350.0281 94 4.9396985 1404.4781 95 4.9874372 1349.2524 96 5.0351759 1399.4384 97 5.0829146 1361.4241 98 5.1306533 1344.5853 99 5.1783920 1356.9955 100 5.2261307 1332.1119 101 5.2738693 1301.5012 102 5.3216080 1326.4846 103 5.3693467 1287.2454 104 5.4170854 1332.2450 105 5.4648241 1229.5584 106 5.5125628 1244.2119 107 5.5603015 1258.0184 108 5.6080402 1254.8713 109 5.6557789 1262.0623 110 5.7035176 1230.1023 111 5.7512563 1233.8537 112 5.7989950 1216.1396 113 5.8467337 1215.8685 114 5.8944724 1211.5171 115 5.9422111 1199.0001 116 5.9899497 1171.9054 117 6.0376884 1181.0994 118 6.0854271 1198.2750 119 6.1331658 1229.1576 120 6.1809045 1166.3230 121 6.2286432 1169.9714 122 6.2763819 1161.8483 123 6.3241206 1146.7722 124 6.3718593 1155.9541 125 6.4195980 1096.9767 126 6.4673367 1138.9220 127 6.5150754 1096.1059 128 6.5628141 1108.8632 129 6.6105528 1134.3928 130 6.6582915 1098.3126 131 6.7060302 1088.4318 132 6.7537688 1079.5525 133 6.8015075 1083.8397 134 6.8492462 1087.4370 135 6.8969849 1072.2796 136 6.9447236 1041.1402 137 6.9924623 1016.1722 138 7.0402010 1070.7651 139 7.0879397 1034.3210 140 7.1356784 1030.6374 141 7.1834171 1032.9864 142 7.2311558 1057.6637 143 7.2788945 1006.9527 144 7.3266332 998.0750 145 7.3743719 1050.5260 146 7.4221106 1001.6985 147 7.4698492 1017.5034 148 7.5175879 1035.8410 149 7.5653266 1000.4575 150 7.6130653 981.2414 151 7.6608040 995.7645 152 7.7085427 990.4387 153 7.7562814 991.1606 154 7.8040201 971.0155 155 7.8517588 998.3966 156 7.8994975 931.0026 157 7.9472362 975.4299 158 7.9949749 927.4493 159 8.0427136 970.2200 160 8.0904523 967.1732 161 8.1381910 945.9498 162 8.1859296 925.6261 163 8.2336683 912.5689 164 8.2814070 886.4218 165 8.3291457 932.2652 166 8.3768844 934.8274 167 8.4246231 896.7846 168 8.4723618 882.6420 169 8.5201005 913.2128 170 8.5678392 889.3674 171 8.6155779 854.7789 172 8.6633166 875.5996 173 8.7110553 836.3349 174 8.7587940 856.6303 175 8.8065327 863.4345 176 8.8542714 866.2489 177 8.9020101 865.2650 178 8.9497487 855.2677 179 8.9974874 854.5842 180 9.0452261 851.7942 181 9.0929648 847.0540 182 9.1407035 823.5921 183 9.1884422 835.0870 184 9.2361809 871.9972 185 9.2839196 829.6416 186 9.3316583 772.8517 187 9.3793970 858.0137 188 9.4271357 847.5978 189 9.4748744 842.0941 190 9.5226131 843.5219 191 9.5703518 849.2675 192 9.6180905 768.8208 193 9.6658291 801.2585 194 9.7135678 817.6215 195 9.7613065 832.8068 196 9.8090452 789.1673 197 9.8567839 790.5499 198 9.9045226 805.7077 199 9.9522613 802.3270 200 10.0000000 714.8252

*layers*layers[[1]] geom_line: na.rm = FALSE, orientation = NA stat_identity: na.rm = FALSE position_identity

$layers[[2]] mapping: xintercept = ~xintercept geom_vline: na.rm = FALSE stat_identity: na.rm = FALSE position_identity

$scales <ggproto object: Class ScalesList, gg> add: function add_defaults: function add_missing: function backtransform_df: function clone: function find: function get_scales: function has_scale: function input: function map_df: function n: function non_position_scales: function scales: list set_palettes: function train_df: function transform_df: function super: <ggproto object: Class ScalesList, gg>

$guides <Guides[0] ggproto object>

*mapping*x expr: ^alpha env: 0x000001f37689a660

$y expr: ^ESS env: 0x000001f37689a660

attr(,"class") [1] "uneval"

$theme list()

$coordinates <ggproto object: Class CoordCartesian, Coord, gg> aspect: function backtransform_range: function clip: on default: TRUE distance: function draw_panel: function expand: TRUE is_free: function is_linear: function labels: function limits: list modify_scales: function range: function render_axis_h: function render_axis_v: function render_bg: function render_fg: function reverse: none setup_data: function setup_layout: function setup_panel_guides: function setup_panel_params: function setup_params: function train_panel_guides: function transform: function super: <ggproto object: Class CoordCartesian, Coord, gg>

$facet <ggproto object: Class FacetNull, Facet, gg> attach_axes: function attach_strips: function compute_layout: function draw_back: function draw_front: function draw_labels: function draw_panel_content: function draw_panels: function finish_data: function format_strip_labels: function init_gtable: function init_scales: function map_data: function params: list set_panel_size: function setup_data: function setup_panel_params: function setup_params: function shrink: TRUE train_scales: function vars: function super: <ggproto object: Class FacetNull, Facet, gg>

$plot_env <environment: 0x000001f37689a660>

$layout <ggproto object: Class Layout, gg> coord: NULL coord_params: list facet: NULL facet_params: list finish_data: function get_scales: function layout: NULL map_position: function panel_params: NULL panel_scales_x: NULL panel_scales_y: NULL render: function render_labels: function reset_scales: function resolve_label: function setup: function setup_panel_guides: function setup_panel_params: function train_position: function super: <ggproto object: Class Layout, gg>

*labels*labels$title [1] "Effective sample size - Log-normal"

*labels*x [1] "alpha"

*labels*y [1] "ESS"

*labels*xintercept [1] "xintercept"

attr(,"class") [1] "gg" "ggplot"

$data alpha ESS 1 0.5000000 1319.3795 2 0.5477387 1912.8477 3 0.5954774 2330.9315 4 0.6432161 2215.5659 5 0.6909548 2687.6971 6 0.7386935 2886.8316 7 0.7864322 3112.2439 8 0.8341709 3244.8015 9 0.8819095 3325.8813 10 0.9296482 3365.2703 11 0.9773869 3434.1865 12 1.0251256 3464.8056 13 1.0728643 3451.2389 14 1.1206030 3451.3353 15 1.1683417 3434.9850 16 1.2160804 3458.0452 17 1.2638191 3404.7329 18 1.3115578 3377.4542 19 1.3592965 3308.6355 20 1.4070352 3301.2697 21 1.4547739 3265.9679 22 1.5025126 3257.6549 23 1.5502513 3193.1753 24 1.5979899 3146.0576 25 1.6457286 3120.5336 26 1.6934673 3029.0822 27 1.7412060 2977.9641 28 1.7889447 2953.1252 29 1.8366834 2911.1499 30 1.8844221 2921.3552 31 1.9321608 2819.7985 32 1.9798995 2794.2426 33 2.0276382 2763.6553 34 2.0753769 2714.7026 35 2.1231156 2699.2433 36 2.1708543 2626.8990 37 2.2185930 2634.9819 38 2.2663317 2570.5997 39 2.3140704 2526.7317 40 2.3618090 2524.3528 41 2.4095477 2488.3205 42 2.4572864 2457.6484 43 2.5050251 2377.9797 44 2.5527638 2378.3955 45 2.6005025 2349.7343 46 2.6482412 2309.2333 47 2.6959799 2270.1100 48 2.7437186 2250.6354 49 2.7914573 2224.0029 50 2.8391960 2181.2765 51 2.8869347 2171.7010 52 2.9346734 2134.9523 53 2.9824121 2119.8229 54 3.0301508 2118.9338 55 3.0778894 2014.2715 56 3.1256281 2041.8941 57 3.1733668 2019.3061 58 3.2211055 2016.2178 59 3.2688442 1972.3623 60 3.3165829 1963.8686 61 3.3643216 1940.6084 62 3.4120603 1871.4359 63 3.4597990 1908.6192 64 3.5075377 1933.0975 65 3.5552764 1861.0219 66

3.6030151 1872.1592 67 3.6507538 1809.7911 68 3.6984925 1805.2188 69 3.7462312 1727.3134 70 3.7939698 1801.9350 71 3.8417085 1741.5585 72 3.8894472 1703.9887 73 3.9371859 1689.9725 74 3.9849246 1697.7065 75 4.0326633 1656.4649 76 4.0804020 1631.3911 77 4.1281407 1623.8876 78 4.1758794 1588.4481 79 4.2236181 1621.9360 80 4.2713568 1632.3783 81 4.3190955 1597.1437 82 4.3668342 1585.1804 83 4.4145729 1504.3002 84 4.4623116 1571.5295 85 4.5100503 1558.3416 86 4.5577889 1512.0101 87 4.6055276 1488.4308 88 4.6532663 1455.2733 89 4.7010050 1505.4202 90 4.7487437 1463.6356 91 4.7964824 1427.3546 92 4.8442211 1450.0974 93 4.8919598 1406.9014 94 4.9396985 1441.2655 95 4.9874372 1435.3170 96 5.0351759 1371.8739 97 5.0829146 1416.5610 98 5.1306533 1361.3420 99 5.1783920 1319.0653 100 5.2261307 1325.5955 101 5.2738693 1344.1213 102 5.3216080 1292.9863 103 5.3693467 1329.4648 104 5.4170854 1329.7286 105 5.4648241 1312.4832 106 5.5125628 1312.5709 107 5.5603015 1278.5675 108 5.6080402 1256.8174 109 5.6557789 1244.2102 110 5.7035176 1267.2343 111 5.7512563 1273.0482 112 5.7989950 1253.6573 113 5.8467337 1216.0213 114 5.8944724 1196.2539 115 5.9422111 1197.1998 116 5.9899497 1206.2816 117 6.0376884 1189.5774 118 6.0854271 1173.1908 119 6.1331658 1194.7551 120 6.1809045 1186.3893 121 6.2286432 1143.7695 122 6.2763819 1154.8908 123 6.3241206 1150.0662 124 6.3718593 1131.8293 125 6.4195980 1131.5125 126 6.4673367 1105.9140 127 6.5150754 1094.0776 128 6.5628141 1108.3928 129 6.6105528 1065.5497 130 6.6582915 1049.5284 131 6.7060302 1074.6158 132 6.7537688 1097.2734 133 6.8015075 1049.4723 134 6.8492462 1065.0721 135 6.8969849 1052.1176 136 6.9447236 1016.0475 137 6.9924623 1004.4967 138 7.0402010 996.6716 139 7.0879397 1049.4997 140 7.1356784 993.9321 141 7.1834171 978.9944 142 7.2311558 1028.4419 143 7.2788945 1006.3743 144 7.3266332 988.8016 145 7.3743719 988.8803 146 7.4221106 974.7751 147 7.4698492 973.9083 148 7.5175879 988.0916 149 7.5653266 1002.1747 150 7.6130653 976.3279 151 7.6608040 968.8756 152 7.7085427 907.8292 153 7.7562814 957.9973 154 7.8040201 936.4883 155 7.8517588 920.5909 156 7.8994975 917.8820 157 7.9472362 908.7540 158 7.9949749 888.4743 159 8.0427136 890.8201 160 8.0904523 902.0726 161 8.1381910 913.9650 162 8.1859296 891.1392 163 8.2336683 890.3638 164 8.2814070 877.8610 165 8.3291457 884.8426 166 8.3768844 899.5071 167 8.4246231 872.7881 168 8.4723618 860.4322 169 8.5201005 896.6338 170 8.5678392 886.7398 171 8.6155779 869.1846 172 8.6633166 826.3664 173 8.7110553 838.1759 174 8.7587940 853.9256 175 8.8065327 818.7317 176 8.8542714 822.1805 177 8.9020101 800.1188 178 8.9497487 830.8864 179 8.9974874 844.1065 180 9.0452261 803.6106 181 9.0929648 818.2383 182 9.1407035 797.3468 183 9.1884422 796.5214 184 9.2361809 814.1617 185 9.2839196 819.0904 186 9.3316583 815.8628 187 9.3793970 814.8775 188 9.4271357 815.6707 189 9.4748744 758.4060 190 9.5226131 808.4794 191 9.5703518 796.4096 192 9.6180905 790.1651 193 9.6658291 784.2695 194 9.7135678 726.4532 195 9.7613065 758.1146 196 9.8090452 741.8261 197 9.8567839 732.5765 198 9.9045226 772.2421 199 9.9522613 740.0048 200 10.0000000 740.0496

*layers*layers[[1]] geom_line: na.rm = FALSE, orientation = NA stat_identity: na.rm = FALSE position_identity

$layers[[2]] mapping: xintercept = ~xintercept geom_vline: na.rm = FALSE stat_identity: na.rm = FALSE position_identity

$scales <ggproto object: Class ScalesList, gg> add: function add_defaults: function add_missing: function backtransform_df: function clone: function find: function get_scales: function has_scale: function input: function map_df: function n: function non_position_scales: function scales: list set_palettes: function train_df: function transform_df: function super: <ggproto object: Class ScalesList, gg>

$guides <Guides[0] ggproto object>

*mapping*x expr: ^alpha env: 0x000001f377a30ee8

$y expr: ^ESS env: 0x000001f377a30ee8

attr(,"class") [1] "uneval"

$theme list()

$coordinates <ggproto object: Class CoordCartesian, Coord, gg> aspect: function backtransform_range: function clip: on default: TRUE distance: function draw_panel: function expand: TRUE is_free: function is_linear: function labels: function limits: list modify_scales: function range: function render_axis_h: function render_axis_v: function render_bg: function render_fg: function reverse: none setup_data: function setup_layout: function setup_panel_guides: function setup_panel_params: function setup_params: function train_panel_guides: function transform: function super: <ggproto object: Class CoordCartesian, Coord, gg>

$facet <ggproto object: Class FacetNull, Facet, gg> attach_axes: function attach_strips: function compute_layout: function draw_back: function draw_front: function draw_labels: function draw_panel_content: function draw_panels: function finish_data: function format_strip_labels: function init_gtable: function init_scales:

function map_data: function params: list set_panel_size: function setup_data: function setup_panel_params: function setup_params: function shrink: TRUE train_scales: function vars: function super: <ggproto object: Class FacetNull, Facet, gg>

$plot_env <environment: 0x000001f377a30ee8>

$layout <ggproto object: Class Layout, gg> coord: NULL coord_params: list facet: NULL facet_params: list finish_data: function get_scales: function layout: NULL map_position: function panel_params: NULL panel_scales_x: NULL panel_scales_y: NULL render: function render_labels: function reset_scales: function resolve_label: function setup: function setup_panel_guides: function setup_panel_params: function train_position: function super: <ggproto object: Class Layout, gg>

*labels*labels$title [1] "Effective sample size - Log-logistic"

*labels*x [1] "alpha"

*labels*y [1] "ESS"

*labels*xintercept [1] "xintercept"

attr(,"class") [1] "gg" "ggplot"

$data alpha ESS 1 0.5000000 585.0098 2 0.5477387 1363.1914 3 0.5954774 1562.4762 4 0.6432161 1841.4450 5 0.6909548 1952.4081 6 0.7386935 2124.5562 7 0.7864322 2166.7255 8 0.8341709 2221.7299 9 0.8819095 2232.4589 10 0.9296482 2171.1740 11 0.9773869 2165.1255 12 1.0251256 2048.4219 13 1.0728643 2002.7542 14 1.1206030 1953.2281 15 1.1683417 1974.1544 16 1.2160804 1896.8220 17 1.2638191 1797.3541 18 1.3115578 1734.2703 19 1.3592965 1720.2447 20 1.4070352 1655.6262 21 1.4547739 1606.8315 22 1.5025126 1567.4774 23 1.5502513 1507.0698 24 1.5979899 1468.2219 25 1.6457286 1399.8278 26 1.6934673 1361.8526 27 1.7412060 1366.8266 28 1.7889447 1288.9761 29 1.8366834 1288.2857 30 1.8844221 1198.9326 31 1.9321608 1174.2053 32 1.9798995 1199.5382 33 2.0276382 1100.3685 34 2.0753769 1060.3157 35 2.1231156 1063.0318 36 2.1708543 997.9937 37 2.2185930 1010.2949 38 2.2663317 971.5292 39 2.3140704 964.6024 40 2.3618090 993.2340 41 2.4095477 909.2050 42 2.4572864 879.7587 43 2.5050251 925.2559 44 2.5527638 850.1816 45 2.6005025 876.1284 46 2.6482412 827.0424 47 2.6959799 792.4361 48 2.7437186 777.9285 49 2.7914573 749.9257 50 2.8391960 747.5844 51 2.8869347 741.8723 52 2.9346734 746.9807 53 2.9824121 680.5196 54 3.0301508 679.8812 55 3.0778894 672.1481 56 3.1256281 659.9956 57 3.1733668 650.9897 58 3.2211055 588.4439 59 3.2688442 610.9740 60 3.3165829 601.6614 61 3.3643216 624.7418 62 3.4120603 586.1596 63 3.4597990 560.5074 64 3.5075377 535.6539 65 3.5552764 561.1755 66 3.6030151 542.8253 67 3.6507538 531.9569 68 3.6984925 551.1323 69 3.7462312 510.7575 70 3.7939698 535.6999 71 3.8417085 477.6326 72 3.8894472 505.8898 73 3.9371859 477.2440 74 3.9849246 463.0654 75 4.0326633 476.2155 76 4.0804020 463.8449 77 4.1281407 444.1749 78 4.1758794 440.0679 79 4.2236181 443.2609 80 4.2713568 457.2297 81 4.3190955 446.2916 82 4.3668342 407.0035 83 4.4145729 437.3858 84 4.4623116 412.1025 85 4.5100503 390.8478 86 4.5577889 393.2495 87 4.6055276 410.8999 88 4.6532663 401.8068 89 4.7010050 363.4365 90 4.7487437 372.0870 91 4.7964824 366.2234 92 4.8442211 345.6600 93 4.8919598 355.9245 94 4.9396985 391.9670 95 4.9874372 359.9705 96 5.0351759 335.6853 97 5.0829146 343.7019 98 5.1306533 352.4243 99 5.1783920 315.4699 100 5.2261307 335.0717 101 5.2738693 347.0030 102 5.3216080 333.8563 103 5.3693467 313.1319 104 5.4170854 312.5391 105 5.4648241 287.3829 106 5.5125628 295.8838 107 5.5603015 303.1226 108 5.6080402 289.4734 109 5.6557789 295.5919 110 5.7035176 271.6495 111 5.7512563 263.7248 112 5.7989950 278.7005 113 5.8467337 296.2085 114 5.8944724 293.7160 115 5.9422111 275.7014 116 5.9899497 269.6878 117 6.0376884 272.3701 118 6.0854271 242.5367 119 6.1331658 280.1751 120 6.1809045 261.8568 121 6.2286432 245.3321 122 6.2763819 257.8160 123 6.3241206 275.7090 124 6.3718593 233.7866 125 6.4195980 231.0712 126 6.4673367 224.1239 127 6.5150754 248.8584 128 6.5628141 233.2778 129 6.6105528 235.6908 130 6.6582915 240.5706 131 6.7060302 218.5020 132 6.7537688 230.2065 133 6.8015075 239.2239 134 6.8492462 226.0242 135 6.8969849 214.4334 136 6.9447236 234.5199 137 6.9924623 223.9066 138 7.0402010 218.5156 139 7.0879397 215.1163 140 7.1356784 234.6064 141 7.1834171 217.9450 142 7.2311558 206.3174 143 7.2788945 207.2148 144 7.3266332 195.1804 145 7.3743719 210.8934 146 7.4221106 202.2492 147 7.4698492 205.1023 148 7.5175879 208.9884 149 7.5653266 187.8635 150 7.6130653 189.1386 151 7.6608040 201.5006 152 7.7085427 183.9623 153 7.7562814 217.3964 154 7.8040201 187.5925 155 7.8517588 189.4361 156 7.8994975 183.7582 157 7.9472362 181.8255 158 7.9949749 182.8284 159 8.0427136 177.0769 160 8.0904523 175.2587 161 8.1381910 169.6042 162 8.1859296 171.0466 163 8.2336683 183.0672 164 8.2814070 175.7368 165 8.3291457 176.1904 166 8.3768844 155.4979 167 8.4246231 155.6365 168 8.4723618 161.9368 169 8.5201005 176.4555 170 8.5678392 158.8401 171 8.6155779 184.9539 172 8.6633166 163.4142 173 8.7110553 171.2130 174 8.7587940 165.7949 175 8.8065327 164.3129 176 8.8542714 160.3453 177 8.9020101 154.4147 178 8.9497487 161.7049 179 8.9974874 159.0038 180 9.0452261 174.8321 181 9.0929648 165.8778 182

9.1407035 149.4150 183 9.1884422 142.5950 184 9.2361809 134.6936 185 9.2839196 142.3531 186 9.3316583 160.7021 187 9.3793970 145.3295 188 9.4271357 147.9348 189 9.4748744 141.2289 190 9.5226131 135.0790 191 9.5703518 140.7623 192 9.6180905 139.8179 193 9.6658291 126.1312 194 9.7135678 137.7435 195 9.7613065 133.9532 196 9.8090452 139.8827 197 9.8567839 147.5589 198 9.9045226 151.6035 199 9.9522613 122.8951 200 10.0000000 135.6446

*layers*layers[[1]] geom_line: na.rm = FALSE, orientation = NA stat_identity: na.rm = FALSE position_identity

$layers[[2]] mapping: xintercept = ~xintercept geom_vline: na.rm = FALSE stat_identity: na.rm = FALSE position_identity

$scales <ggproto object: Class ScalesList, gg> add: function add_defaults: function add_missing: function backtransform_df: function clone: function find: function get_scales: function has_scale: function input: function map_df: function n: function non_position_scales: function scales: list set_palettes: function train_df: function transform_df: function super: <ggproto object: Class ScalesList, gg>

$guides <Guides[0] ggproto object>

*mapping*x expr: ^alpha env: 0x000001f377d79cb8

$y expr: ^ESS env: 0x000001f377d79cb8

attr(,"class") [1] "uneval"

$theme list()

$coordinates <ggproto object: Class CoordCartesian, Coord, gg> aspect: function backtransform_range: function clip: on default: TRUE distance: function draw_panel: function expand: TRUE is_free: function is_linear: function labels: function limits: list modify_scales: function range: function render_axis_h: function render_axis_v: function render_bg: function render_fg: function reverse: none setup_data: function setup_layout: function setup_panel_guides: function setup_panel_params: function setup_params: function train_panel_guides: function transform: function super: <ggproto object: Class CoordCartesian, Coord, gg>

$facet <ggproto object: Class FacetNull, Facet, gg> attach_axes: function attach_strips: function compute_layout: function draw_back: function draw_front: function draw_labels: function draw_panel_content: function draw_panels: function finish_data: function format_strip_labels: function init_gtable: function init_scales: function map_data: function params: list set_panel_size: function setup_data: function setup_panel_params: function setup_params: function shrink: TRUE train_scales: function vars: function super: <ggproto object: Class FacetNull, Facet, gg>

$plot_env <environment: 0x000001f377d79cb8>

$layout <ggproto object: Class Layout, gg> coord: NULL coord_params: list facet: NULL facet_params: list finish_data: function get_scales: function layout: NULL map_position: function panel_params: NULL panel_scales_x: NULL panel_scales_y: NULL render: function render_labels: function reset_scales: function resolve_label: function setup: function setup_panel_guides: function setup_panel_params: function train_position: function super: <ggproto object: Class Layout, gg>

*labels*labels$title [1] "Effective sample size - Gen. Gamma"

*labels*x [1] "alpha"

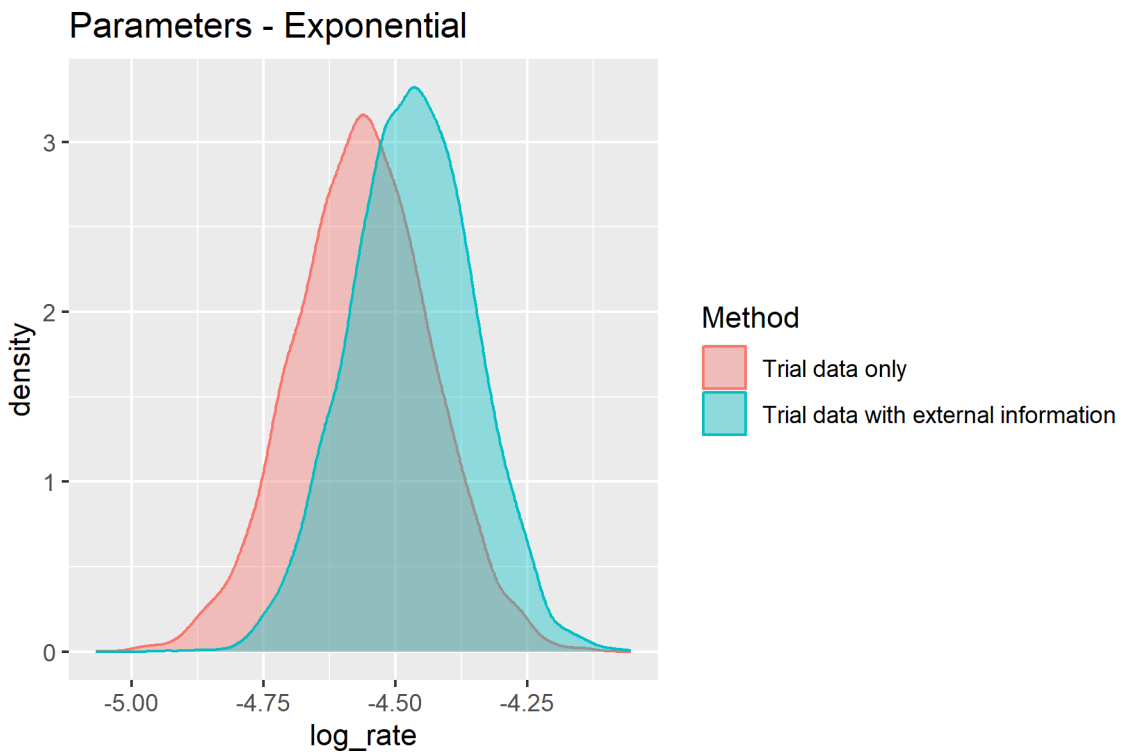
*labels*y [1] "ESS"

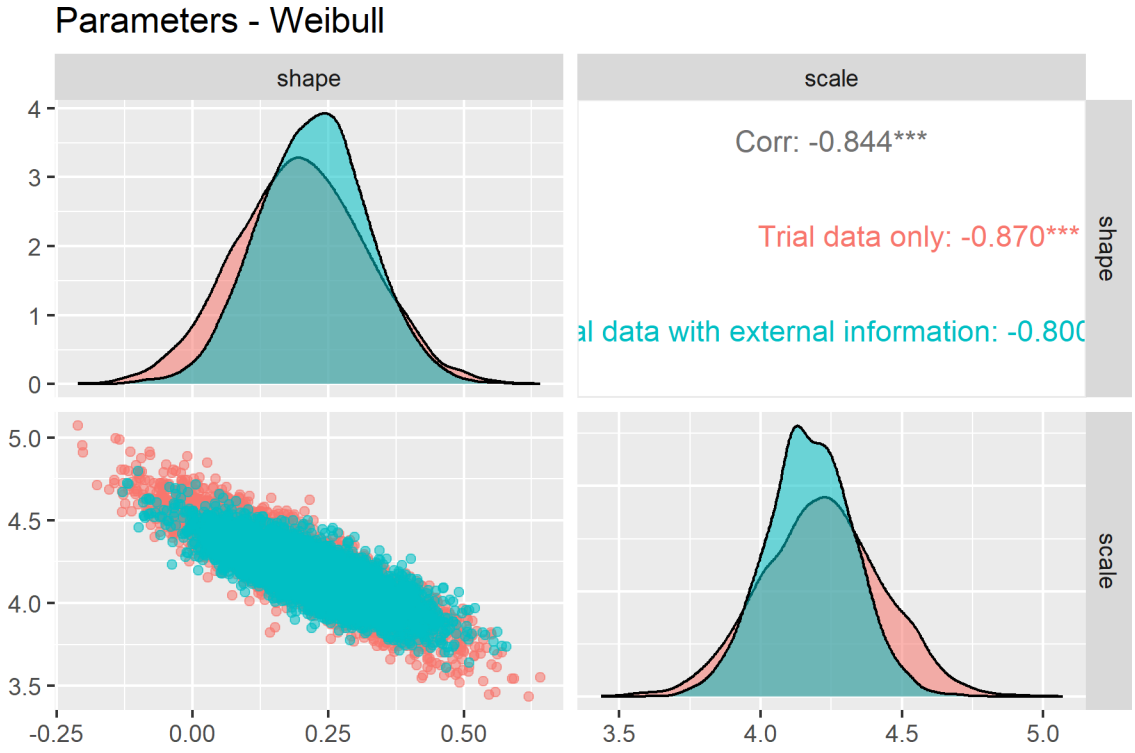
*labels*xintercept [1] "xintercept"

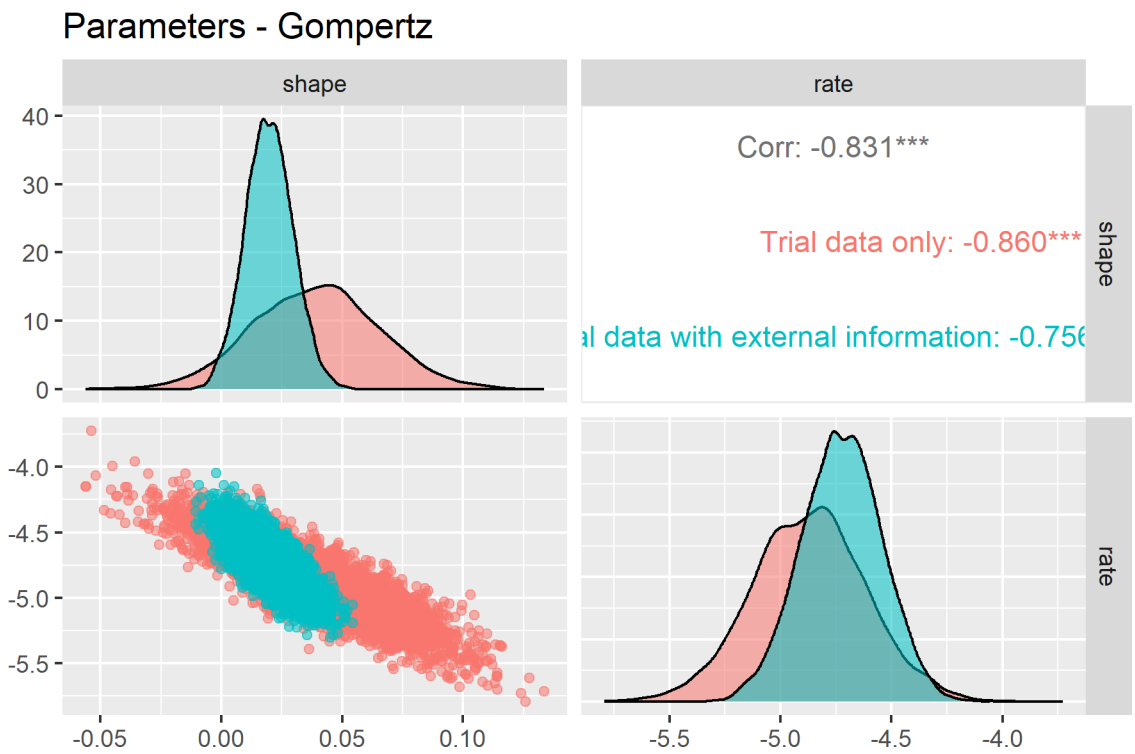
attr(,"class") [1] "gg" "ggplot"

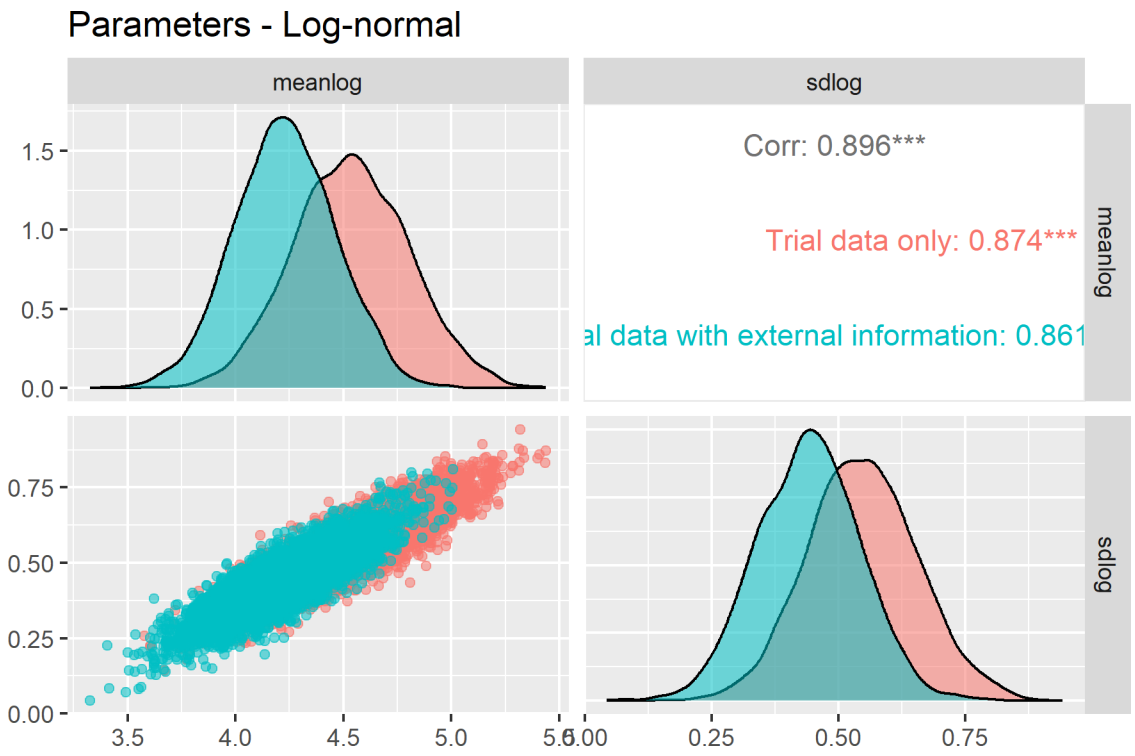
1.3 Distribution comparison plots

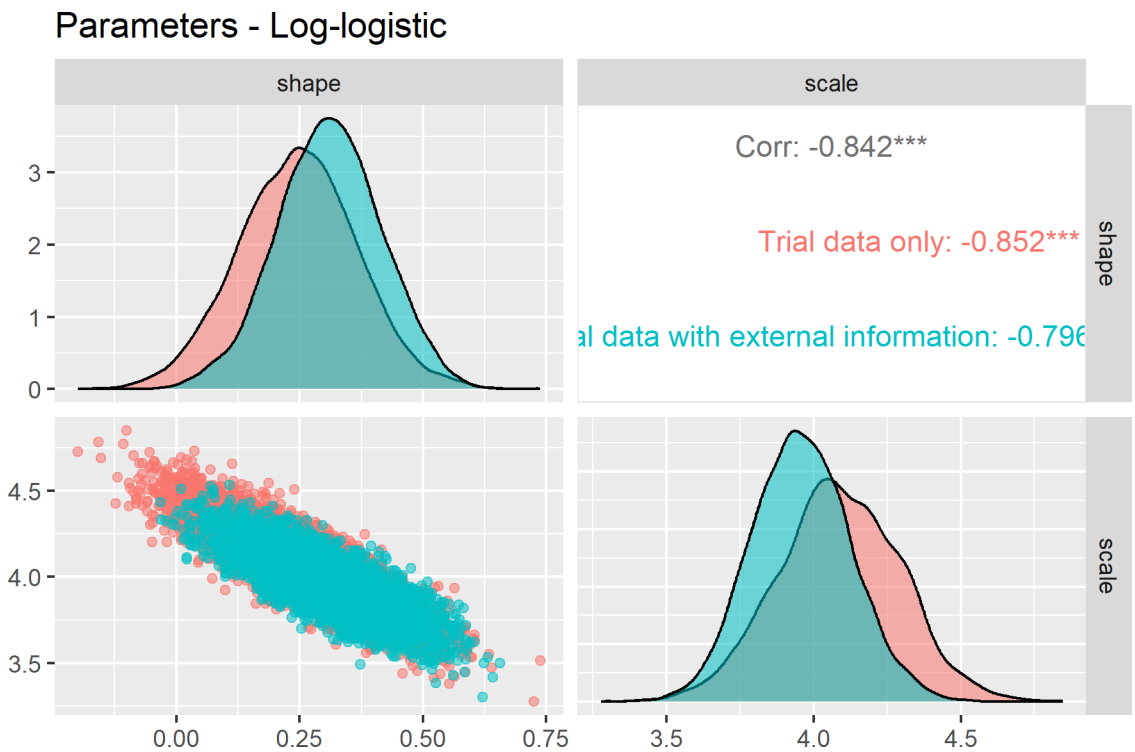
The below plots show the estimated effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the survival distributions under consideration. The vertical line shows the optimal value of alpha identified by the algorithm. Larger ESS values are better.

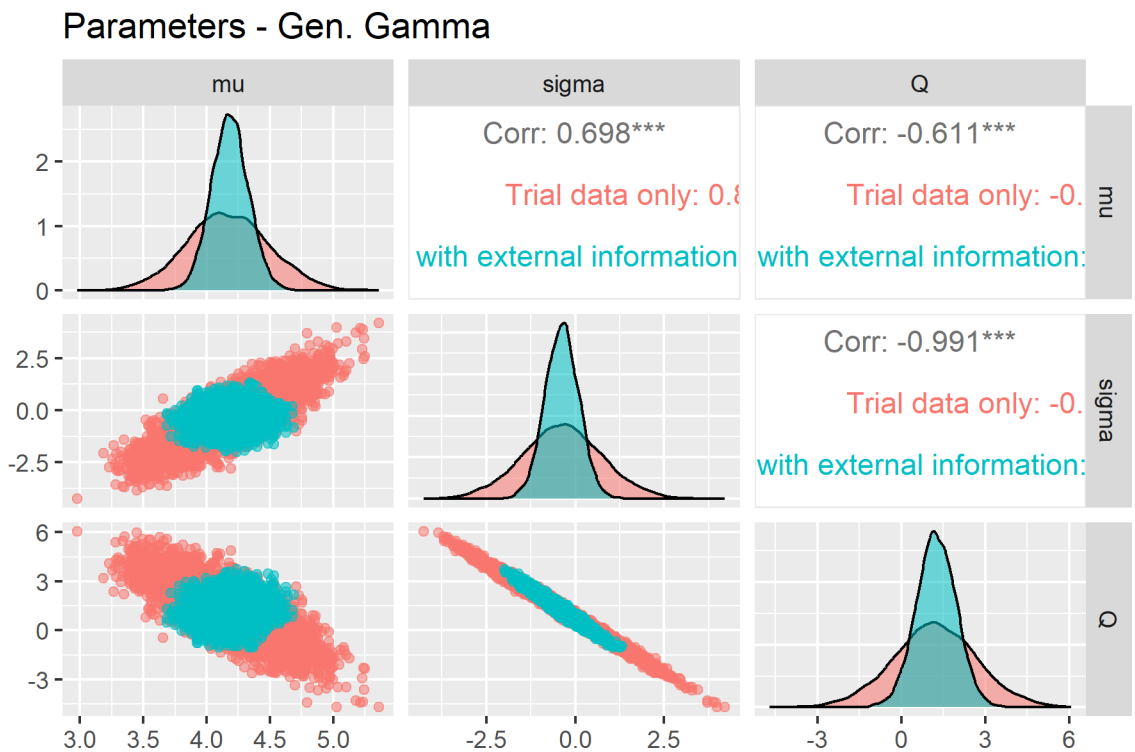

Parameters - Exponential
density
log_rate
-5.00
-4.75
-4.50
-4.25
0
1
2
3
Method
Trial data only
Trial data with external information


Parameters - Weibull
shape
scale
Corr: -0.844***
Trial data only: -0.870***
al data with external information: -0.800
shape
scale
-0.25
0.00
0.25
0.50
3.5
4.0
4.5
5.0


Parameters - Gompertz
shape
rate
Corr: -0.831***
Trial data only: -0.860***
al data with external information: -0.756
shape
rate
40
30
20
10
0
-4.0
-4.5
-5.0
-5.5
-0.05
0.00
0.05
0.10
-5.5
-5.0
-4.5
-4.0

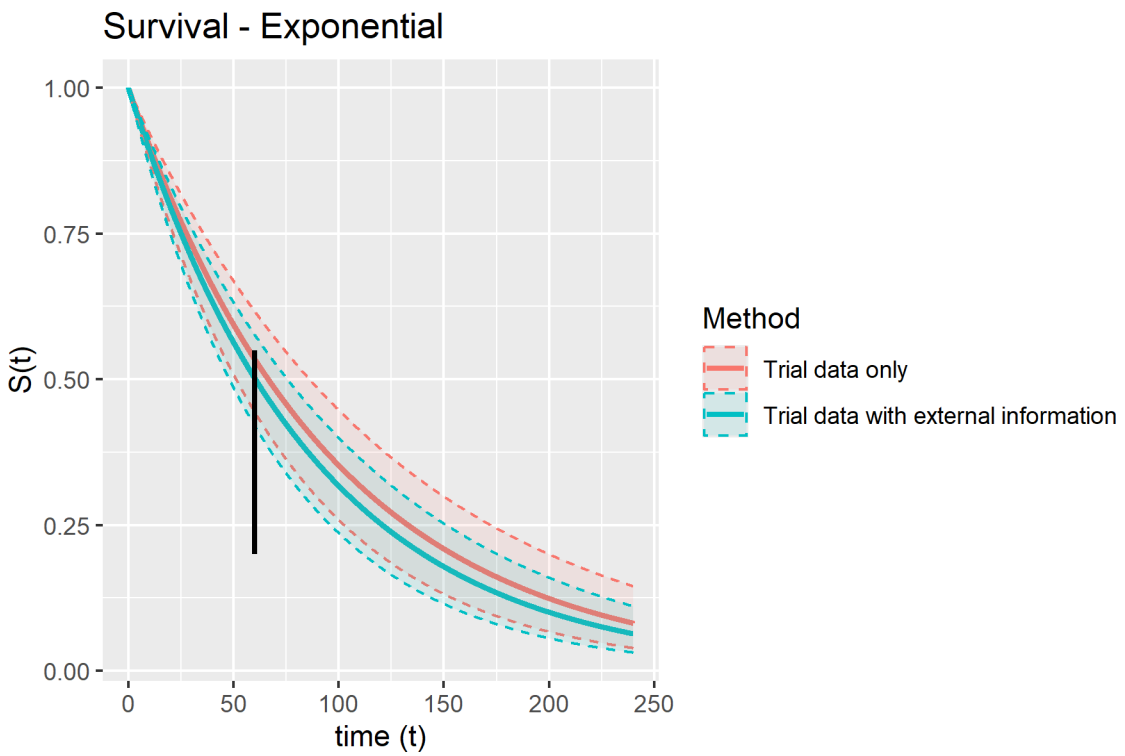

Parameters - Log-normal
meanlog
sdlog
Corr: 0.896***
Trial data only: 0.874***
al data with external information: 0.861
meanlog
sdlog
1.5
1.0
0.5
0.0
0.75
0.50
0.25
0.00
3.5
4.0
4.5
5.0
5.5
0.00
0.25
0.50
0.75

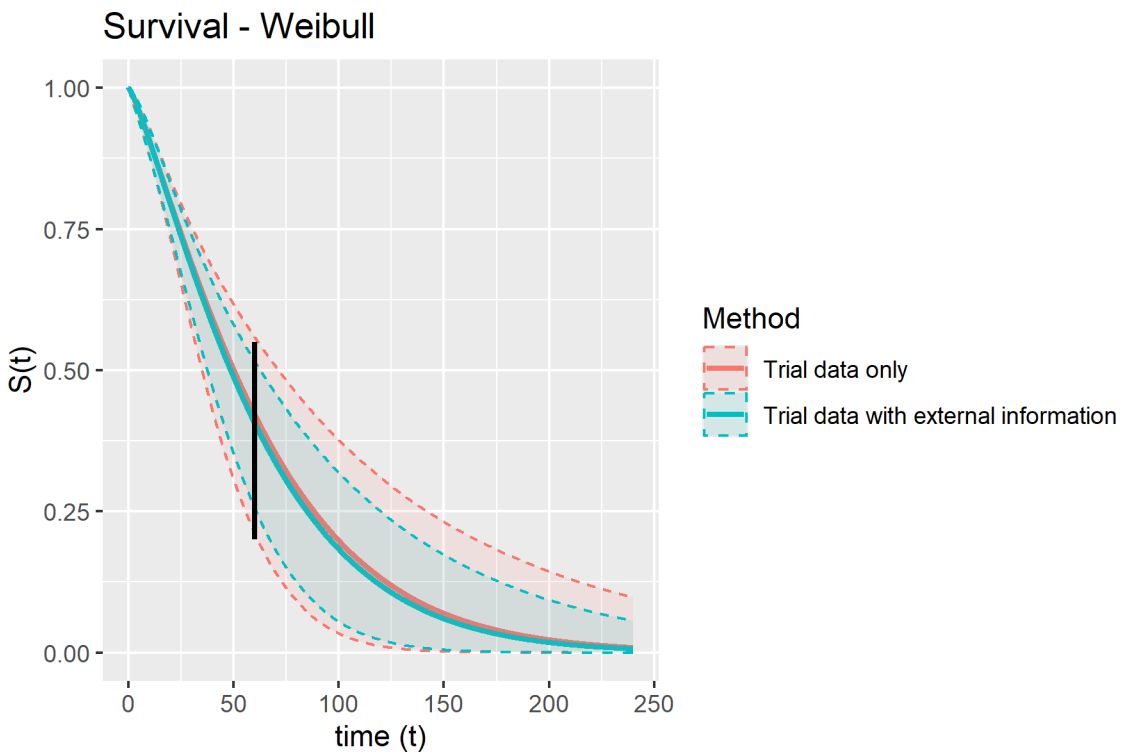


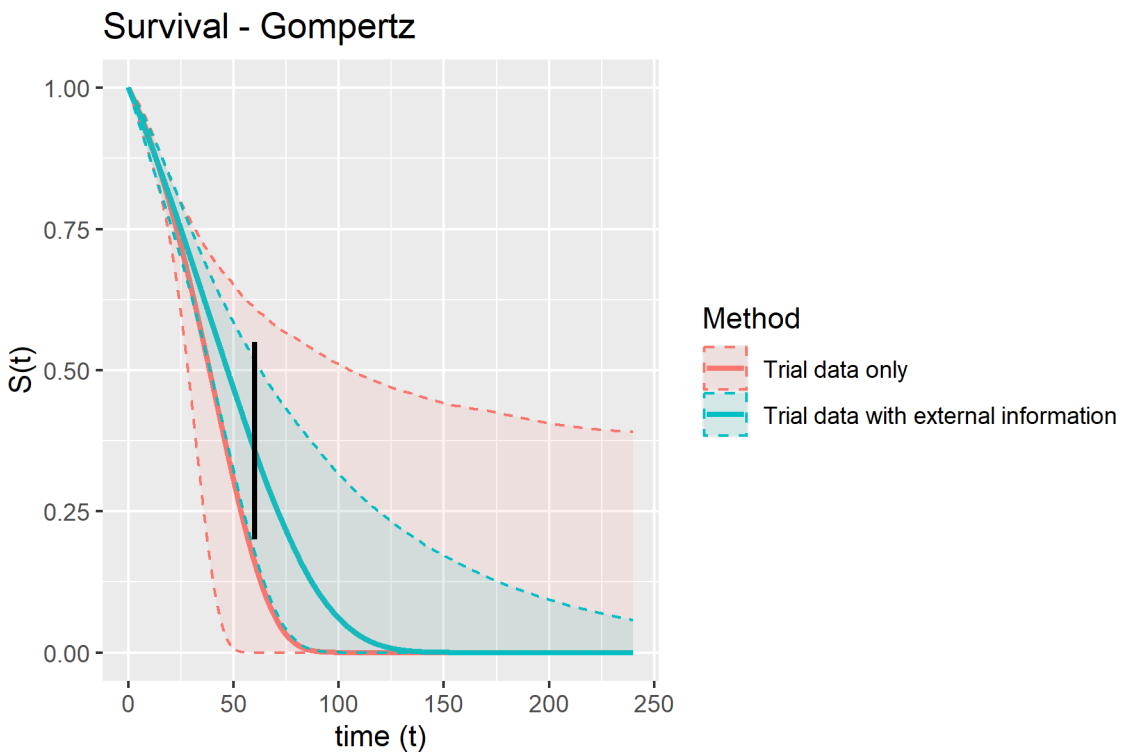


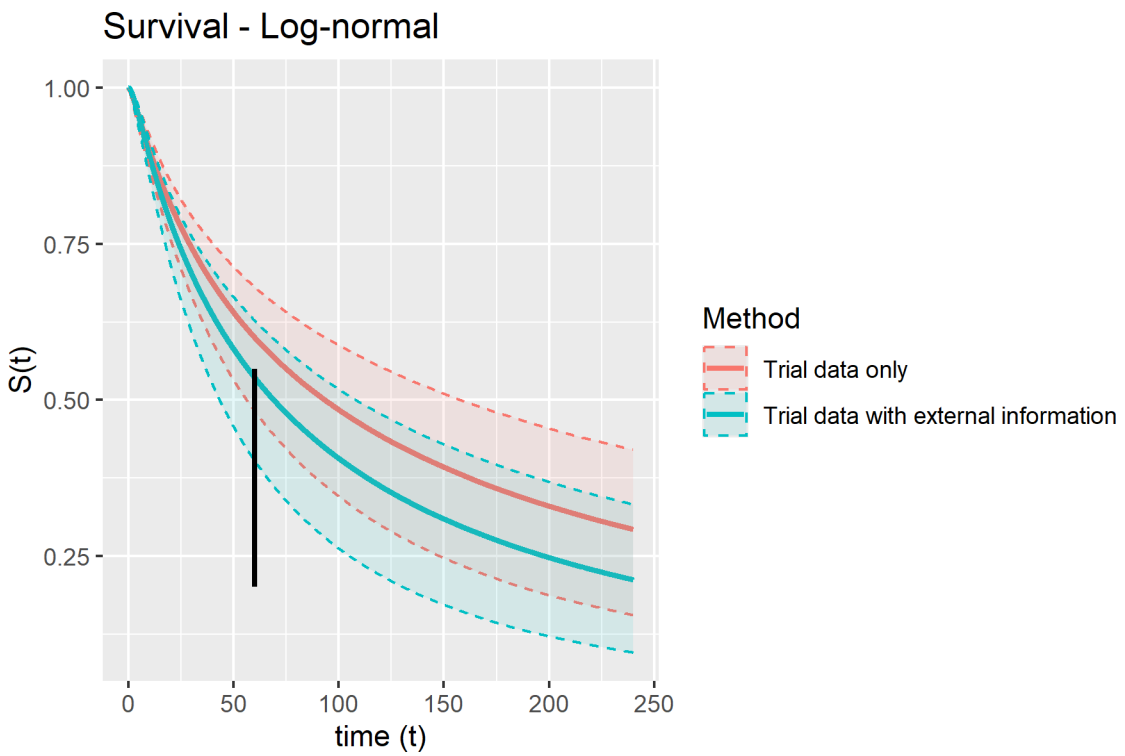
1.4 Survival comparison plots

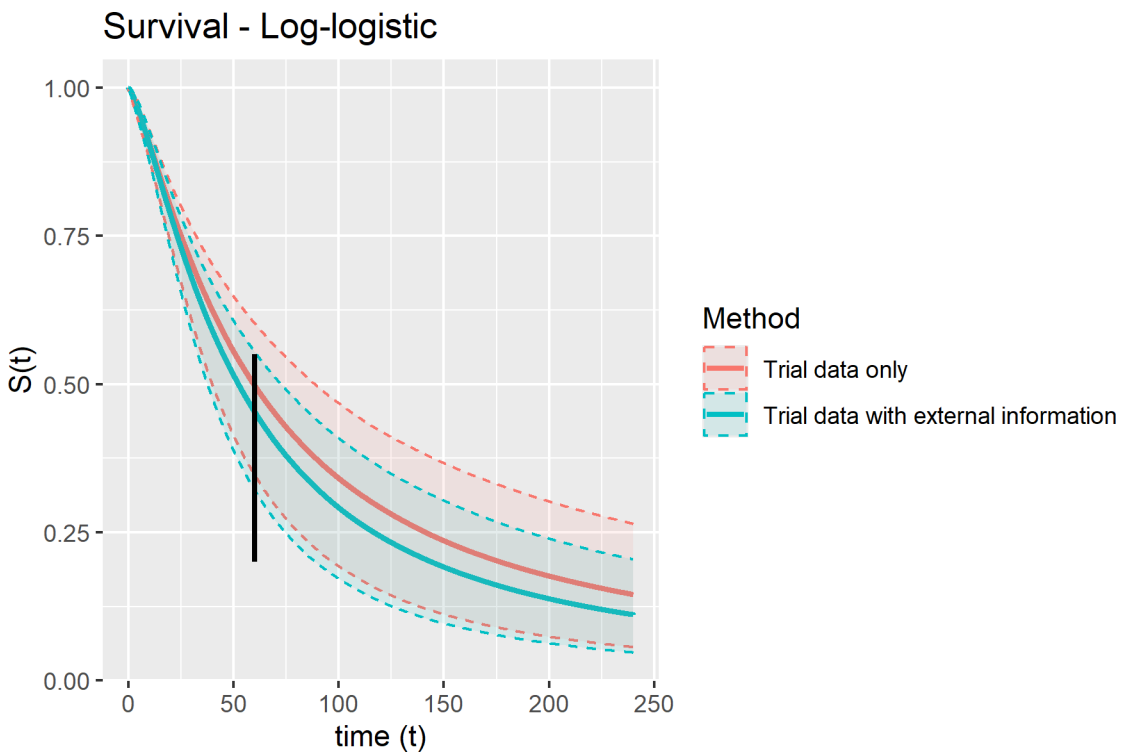
The below plots show the estimated survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, for the survival distributions under consideration. Vertical bars indicate the 95% probability interval for 5 year OS based on external information.

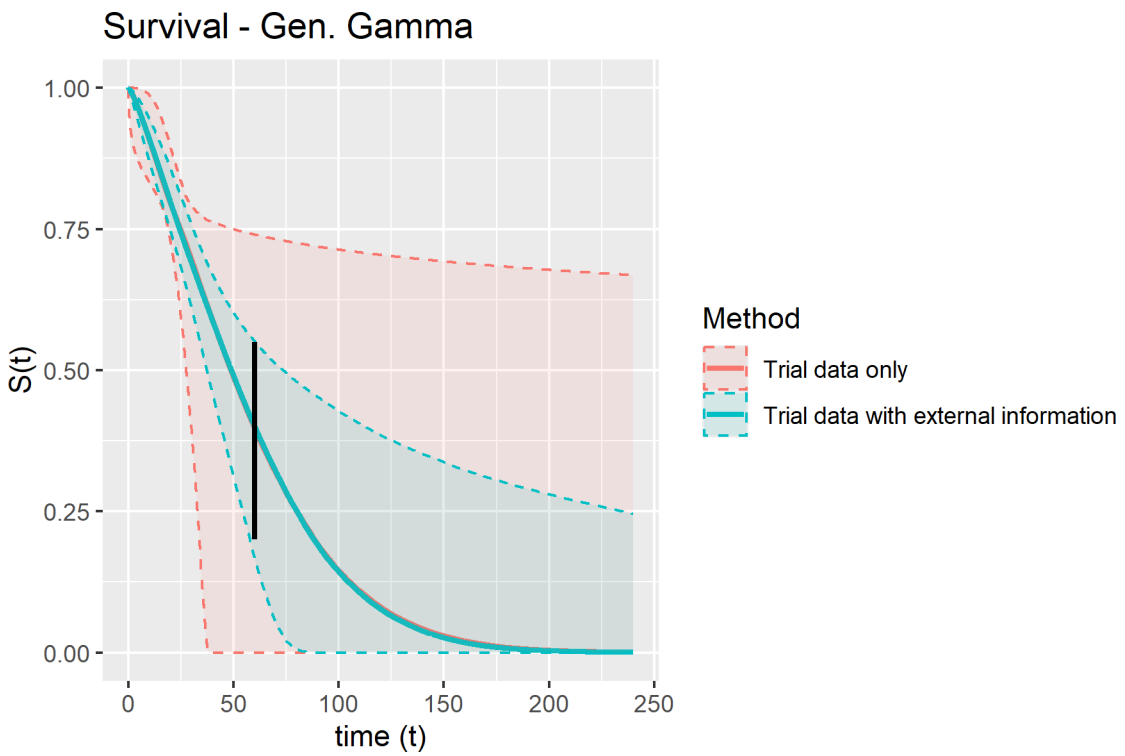

Survival - Exponential
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Weibull
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Gompertz
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Log-normal
1.00
0.75
0.50
0.25
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Log-logistic
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Gen. Gamma
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information

1.5 Parameter variability

Table 13 shows 'Generalised variance,' i.e., determinants of variance-covariance matrices, of the parameter estimates obtained using trial data only (MLE) and combining trial data withe external information (IS). Formally, the generalised variance gives the area of the 95% highest density ellipse and can be interpreted as a 1-parameter measure of parameter uncertainty.

Table 13: Generalised variance of model parameter distributions. Comparison between parameter estimates obtained using trial data only (MLE) and combining trial data with external information (IS).

Distribution	Distance	MLE.Variance	IS.Variance	Var..Ratio
exponential	0.0957855	0.0169492	0.0138444	0.8168192
weibull	0.0413797	0.0001670	0.0000827	0.4948562
gompertz	0.1474560	0.0000114	0.0000013	0.1140736
lognormal	0.3260830	0.0001878	0.0001435	0.7641277
llogis	0.1461999	0.0001754	0.0001134	0.6466361
gengamma	0.0546778	0.0003562	0.0000405	0.1138456

2 Comparison with expertsurv output

2.1 Plots of survival curves over time

Figures 1 to **??** show comparisons of survival curves (median and 95% confidence/credible intervals) obtained from the importance sampling method described in the paper, with those obtained from the fully Bayesian approach of Cooney & White implemented in the expertsurv package.

Survival curves for the exponential, Weibull, Gompertz and log-logistic models were almost identical between the two methods. For the log-normal and generalised gamma models (Figures 4 and **??**), there was a small but notable shift in the parameter distribution and corresponding survival estimates. The reason for this is not immediately clear. However, in both cases the impact was small and mostly evident in the tails of the survival distributions.

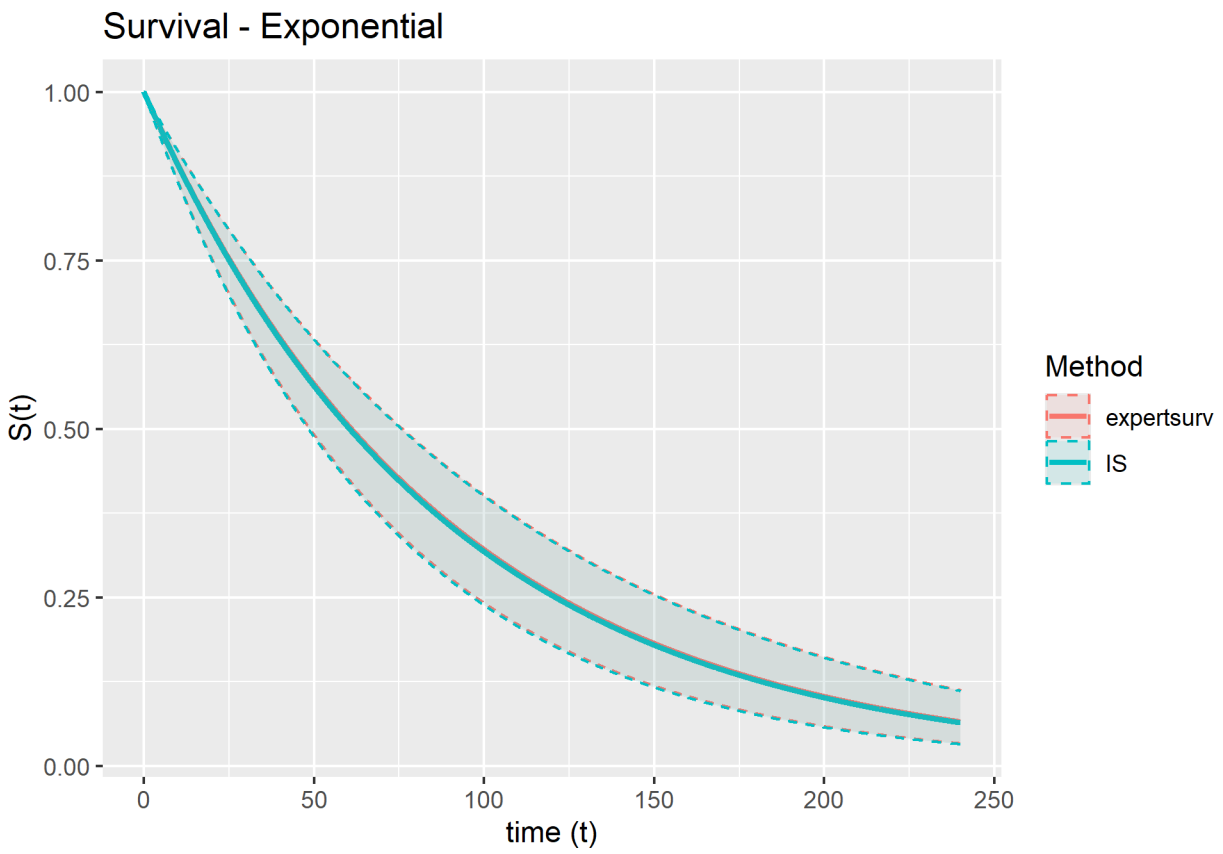


Figure 1: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Exponential distribution

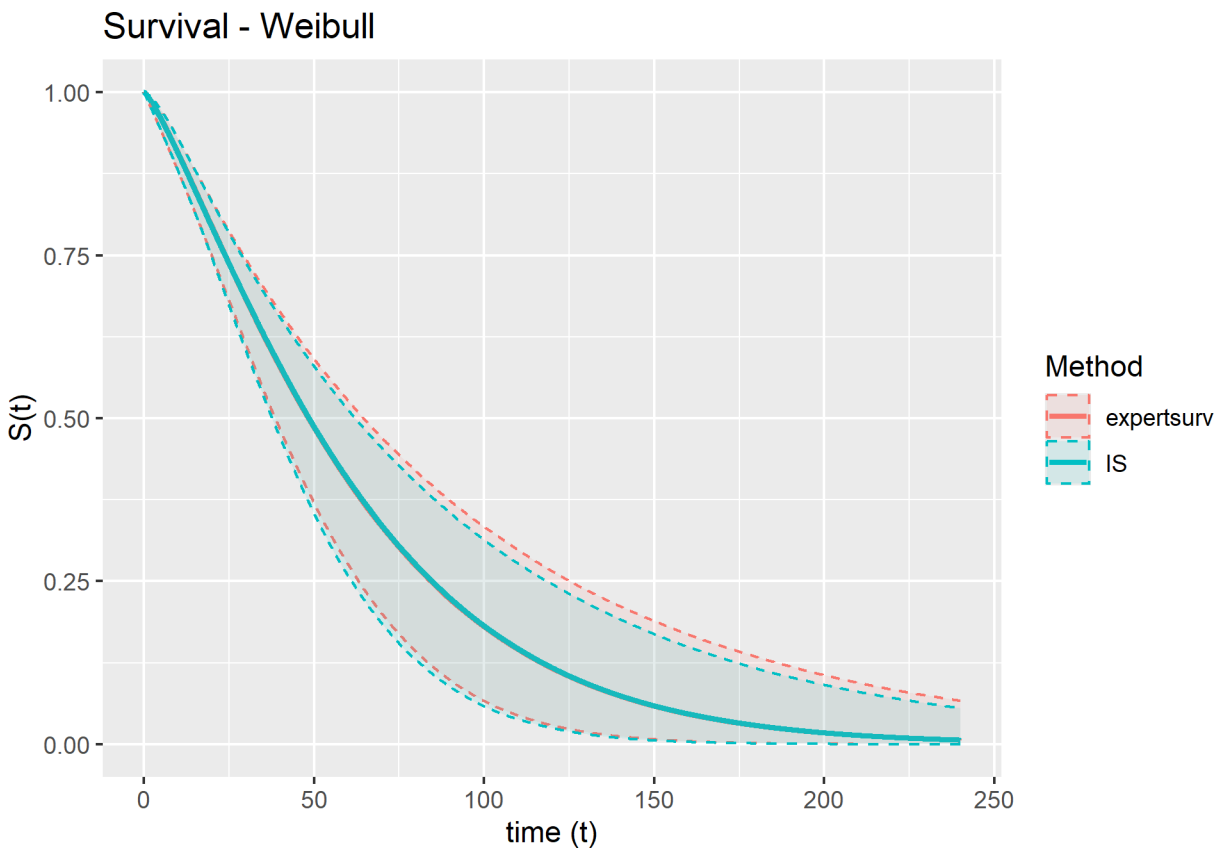


Figure 2: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Weibull distribution

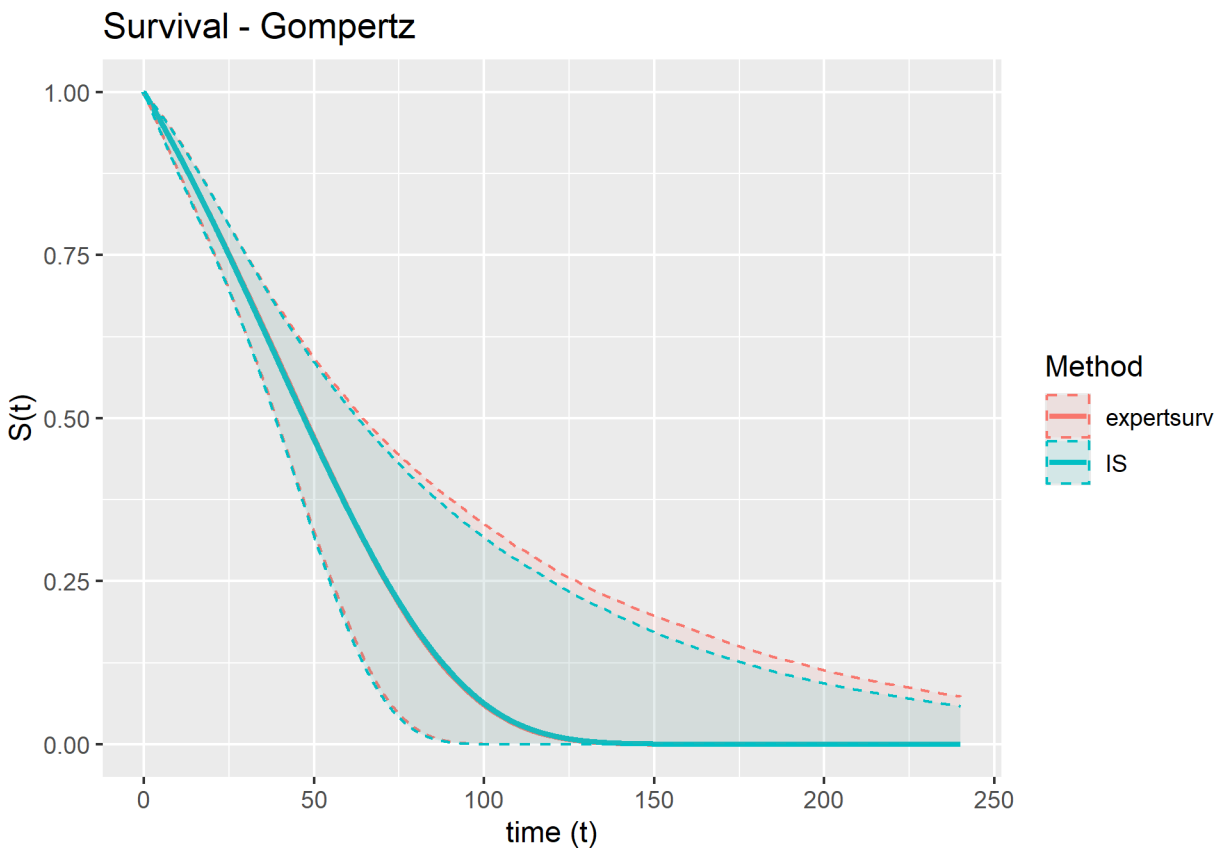


Figure 3: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-normal distribution

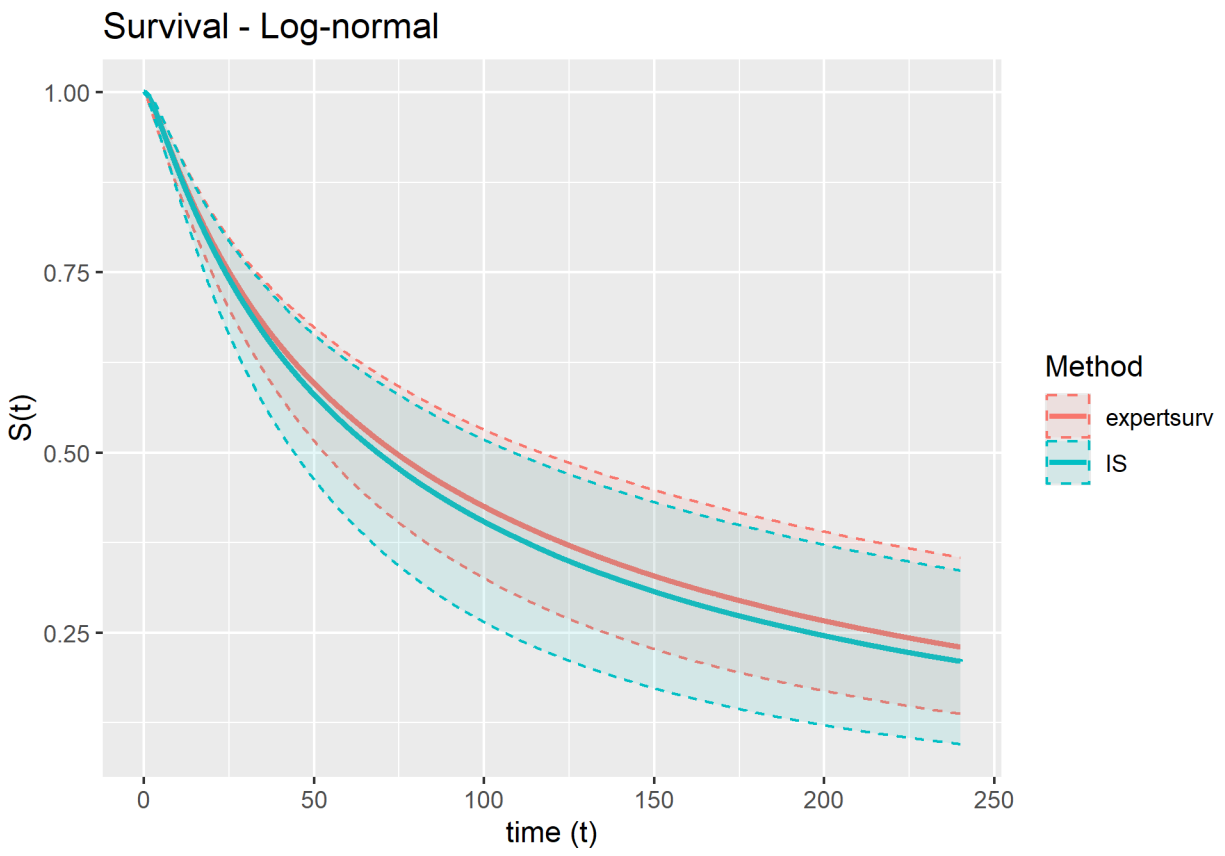


Figure 4: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-logistic distribution

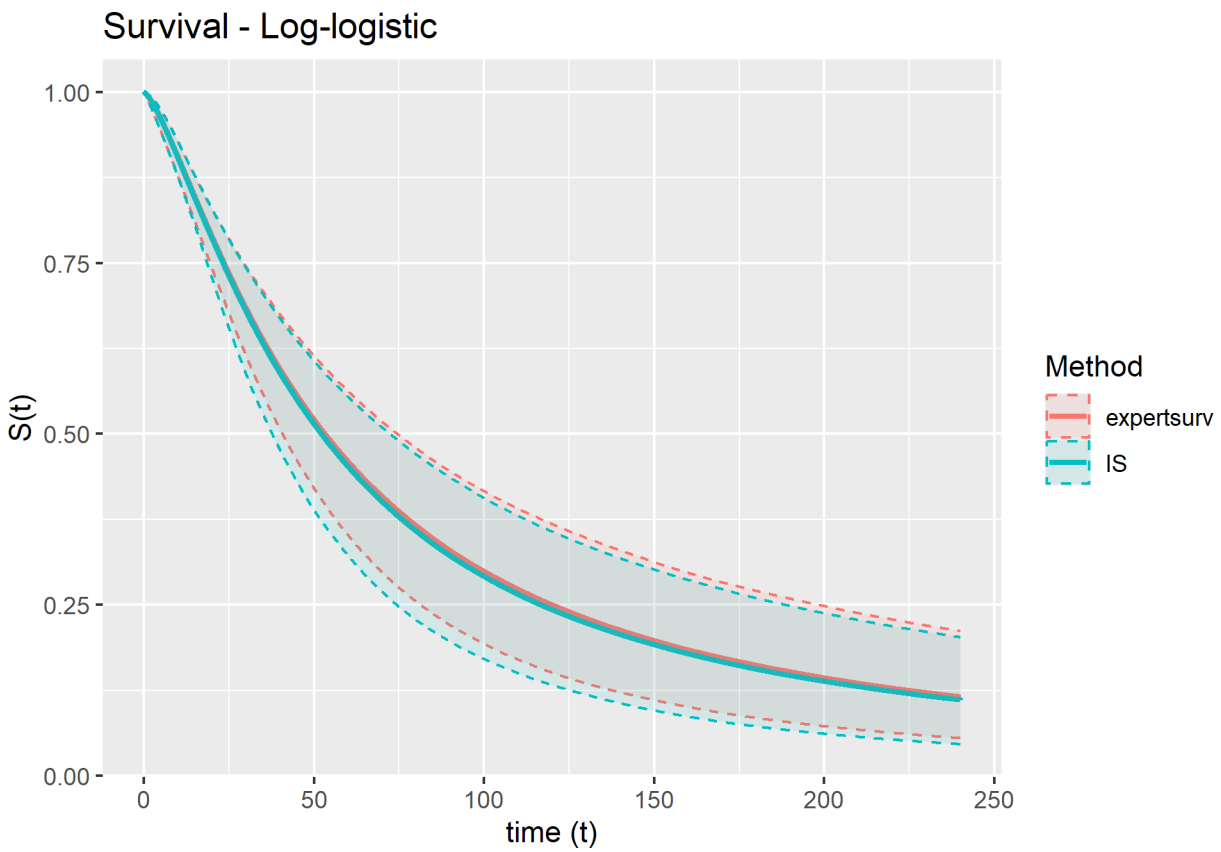


Figure 5: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Gompertz distribution

2.1.1 Comparisons of parameter distributions

Figures 6 to **??** compare survival curve parameter distributions obtained from the importance sampling method described in the paper, with those obtained from the fully Bayesian approach of Cooney & Whilte implemented in the expertsurv package. Again, some differences are evidence between the two methods for the log-normal and generalised gamma models (Figures 9 and **??**).

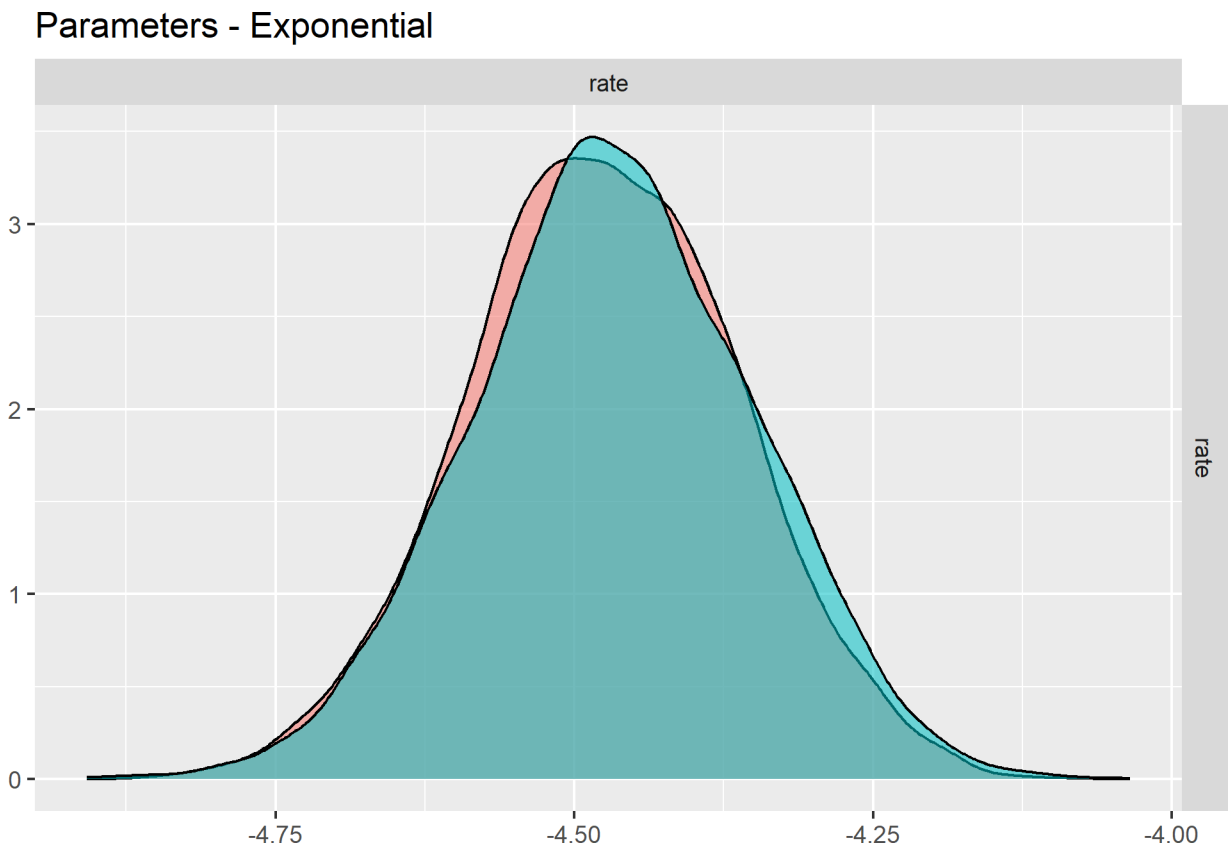


Figure 6: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Exponential distribution

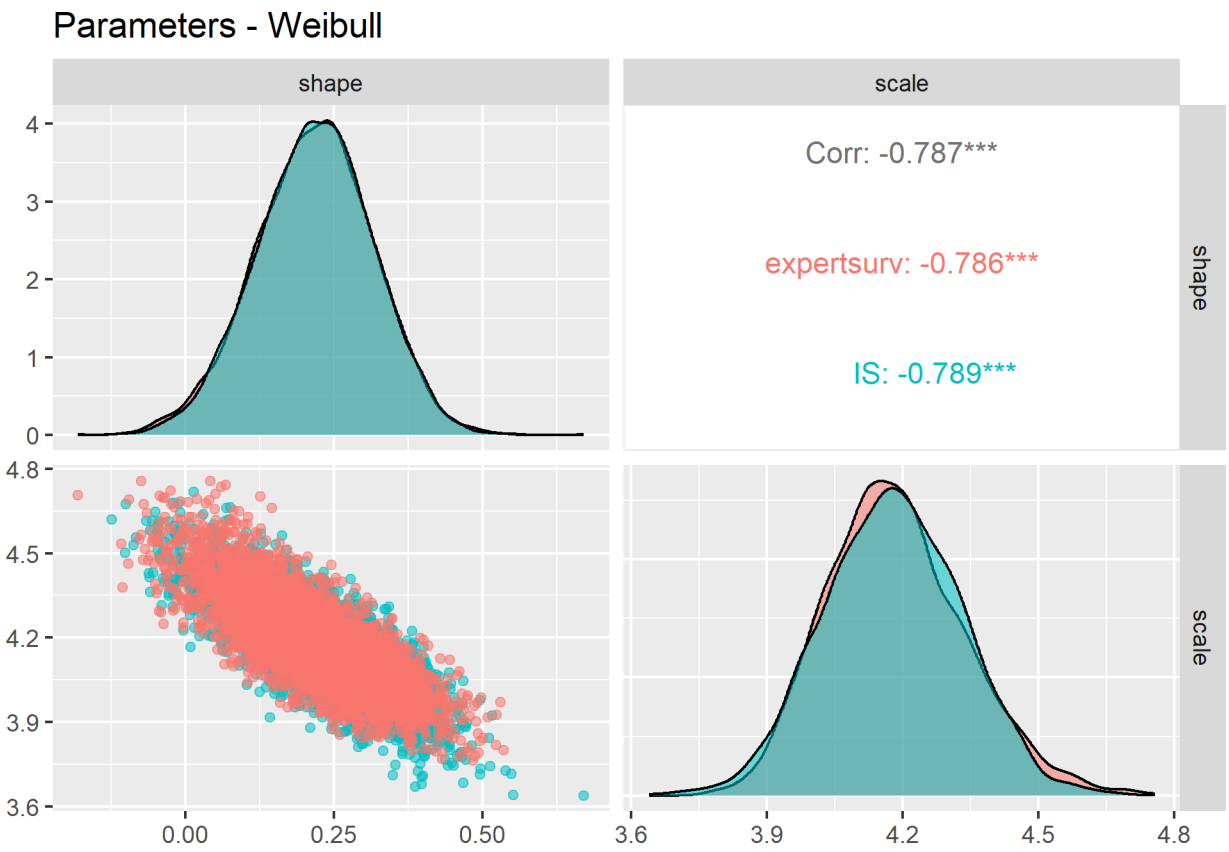


Figure 7: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Weibull distribution

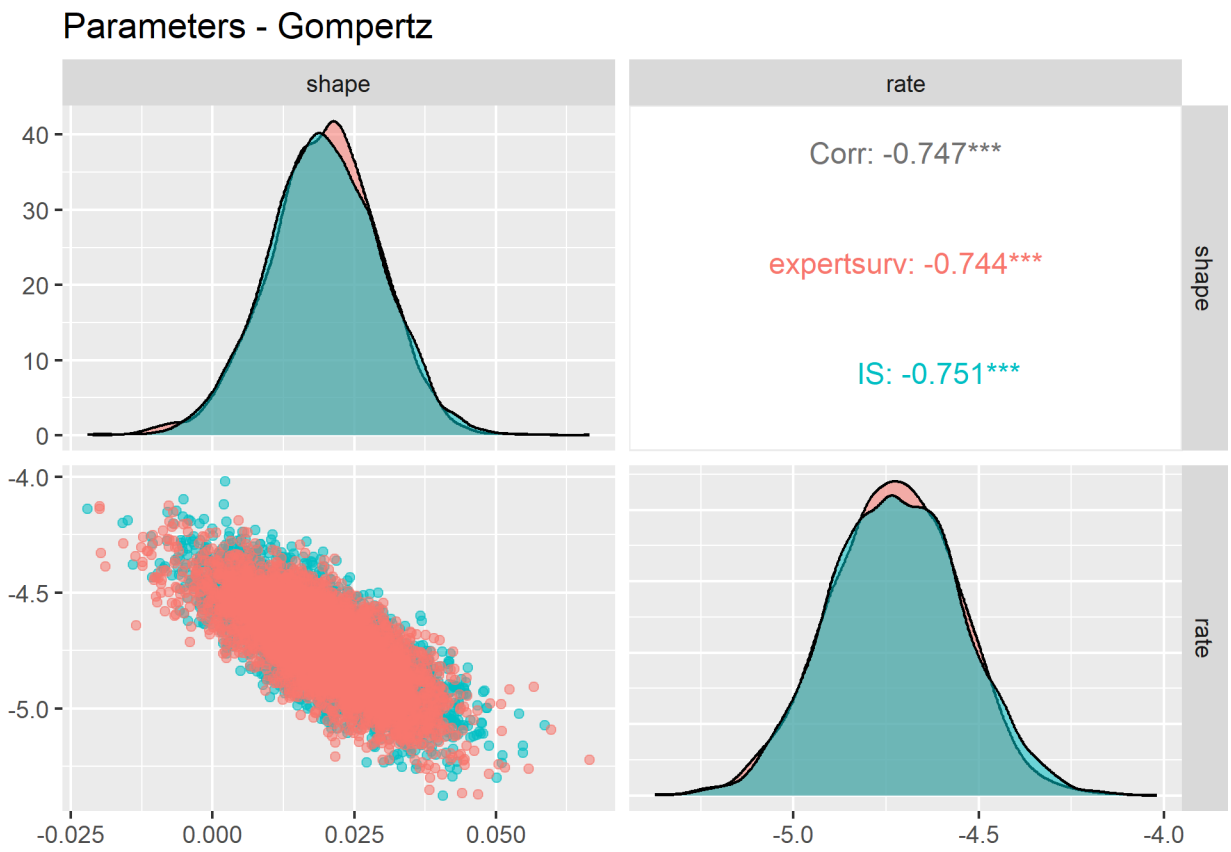


Figure 8: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-normal distribution

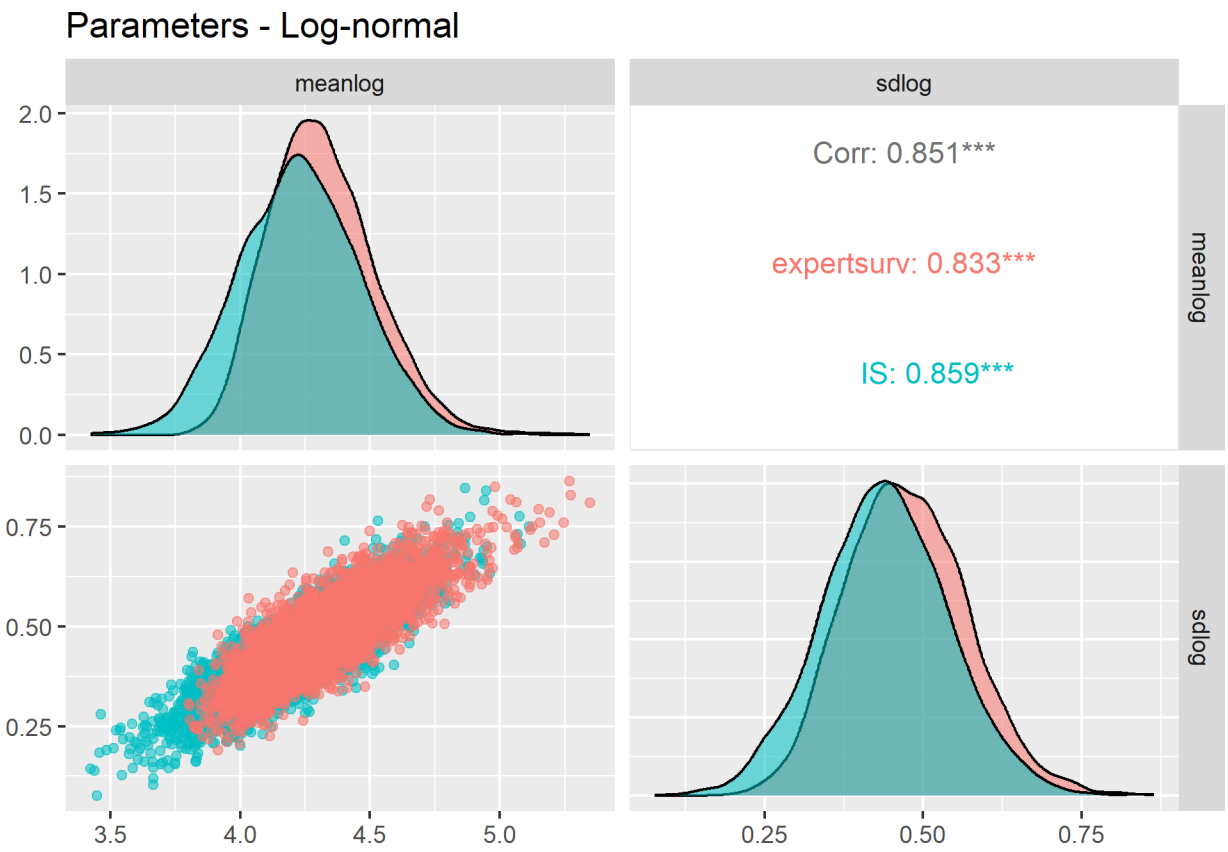


Figure 9: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-logistic distribution

Table 14: Comparison of probabilistic total area under the curve (mean lifetime OS) estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, and with the fully Bayesian method of Cooney and White implemented in the expertsurv package. Values shown are probabilstic mean and 95 percent confidence/credible intervals.

Distribution	expertsurv	IS	Mean.Diff	Var.Ratio
Exponential	88.48 (70.48, 109.73)	87.74 (69.78, 109.28)	-0.74	1.06
Gompertz	55.85 (39.56, 92.84)	53.89 (39.24, 85.24)	-1.96	0.44
Log-logistic	123.17 (77.65, 189.59)	119.06 (71.11, 181.96)	-4.11	1.01
Log-normal	202.54 (126.08, 311.84)	183.80 (97.93, 295.22)	-18.74	1.07
Weibull	62.43 (45.73, 90.16)	61.93 (44.18, 85.17)	-0.50	0.90

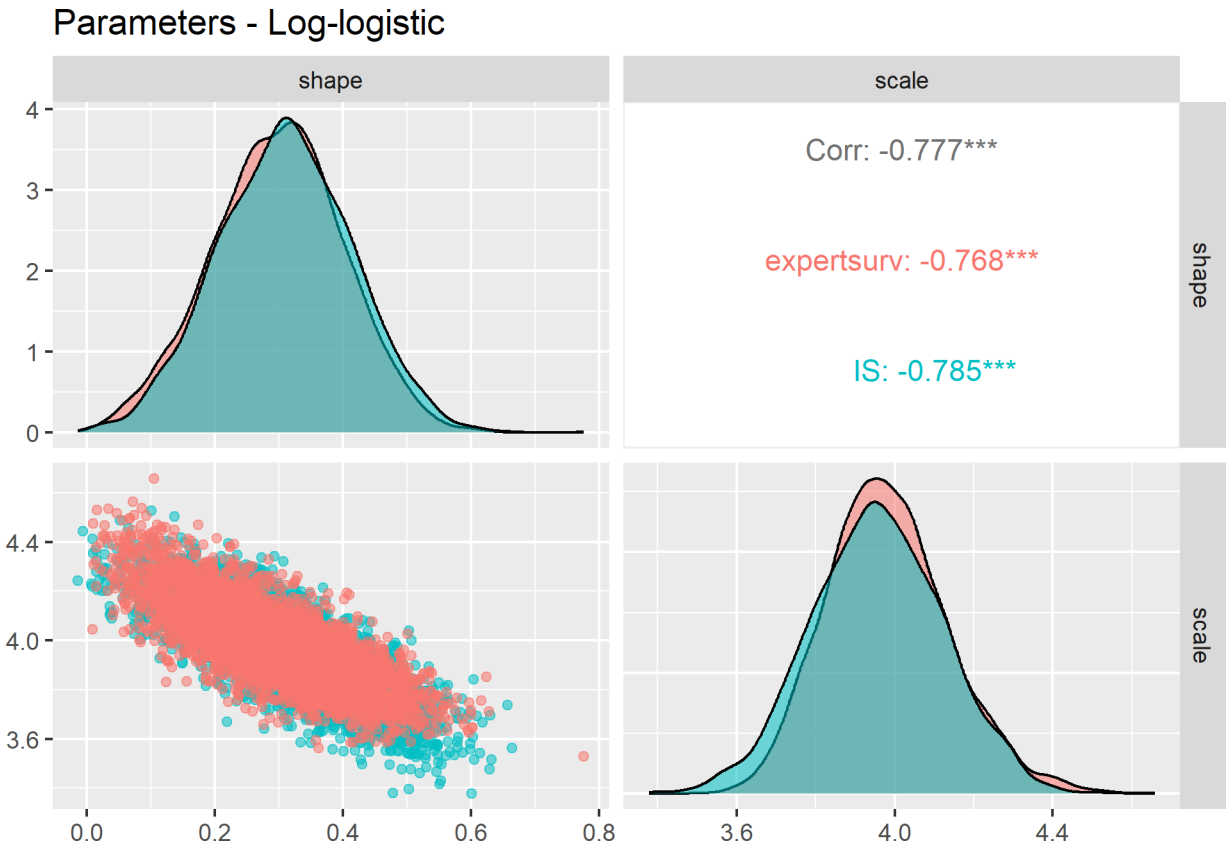


Figure 10: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Gompertz distribution

2.2 Comparisons of AUC distributions

Table 14 and Figure 11 compare probabilistic estimates of area under the curve between the two methods. In general, AUC measures (population) mean survival extrapolated over a lifetime, though in these examples survival is capped at 100 years since some curves plateau.

Table 15: Comparison of 5-year OS estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with the fully Bayesian method of Cooney and White implemented in the expertsurv package.

Distribution	Output	mean	median	sd	lwr.95	upr.95
Exponential	IS	50.16%	50.28%	4.03%	42.32%	57.75%
Exponential	expertsurv	50.47%	50.56%	3.87%	42.68%	57.88%
Gompertz	IS	35.63%	35.98%	8.87%	17.68%	51.87%
Gompertz	expertsurv	35.89%	36.01%	8.71%	18.84%	52.77%
Log-logistic	IS	44.89%	45.25%	5.98%	32.21%	55.46%
Log-logistic	expertsurv	45.88%	45.95%	5.37%	35.19%	56.30%
Log-normal	IS	52.94%	53.44%	5.72%	40.77%	62.69%
Log-normal	expertsurv	55.11%	55.20%	4.50%	46.42%	63.67%
Weibull	IS	40.05%	40.56%	6.65%	25.86%	51.30%
Weibull	expertsurv	40.32%	40.41%	6.53%	27.64%	52.69%

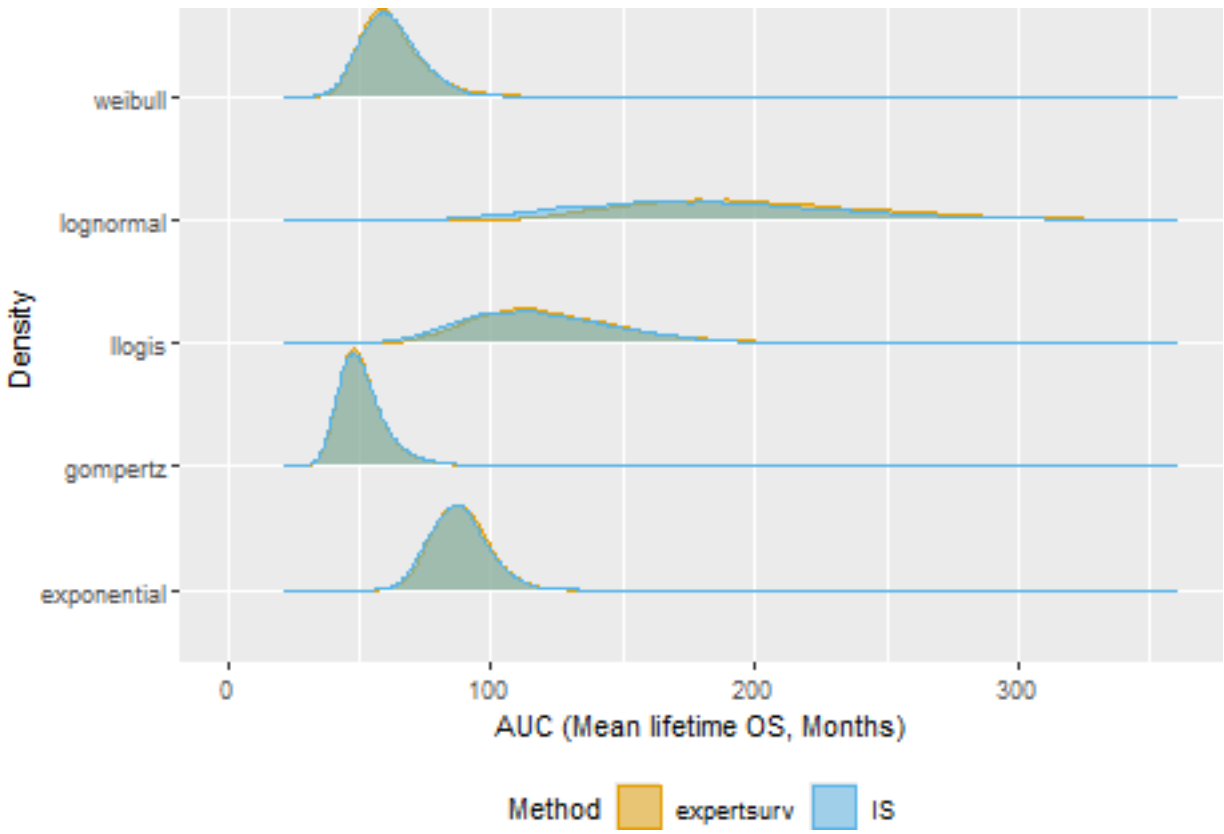


Figure 11: Density plots comparing probabilistic total area under the curve (mean OS) distributions obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with those obtained from the fully Bayesian method of Cooney and White implemented in the expertsurv package.

2.3 Comparisons of 5-year OS

Table 15 and Figure 12 compare probabilistic estimates of 5-year OS obtained from the two methods.

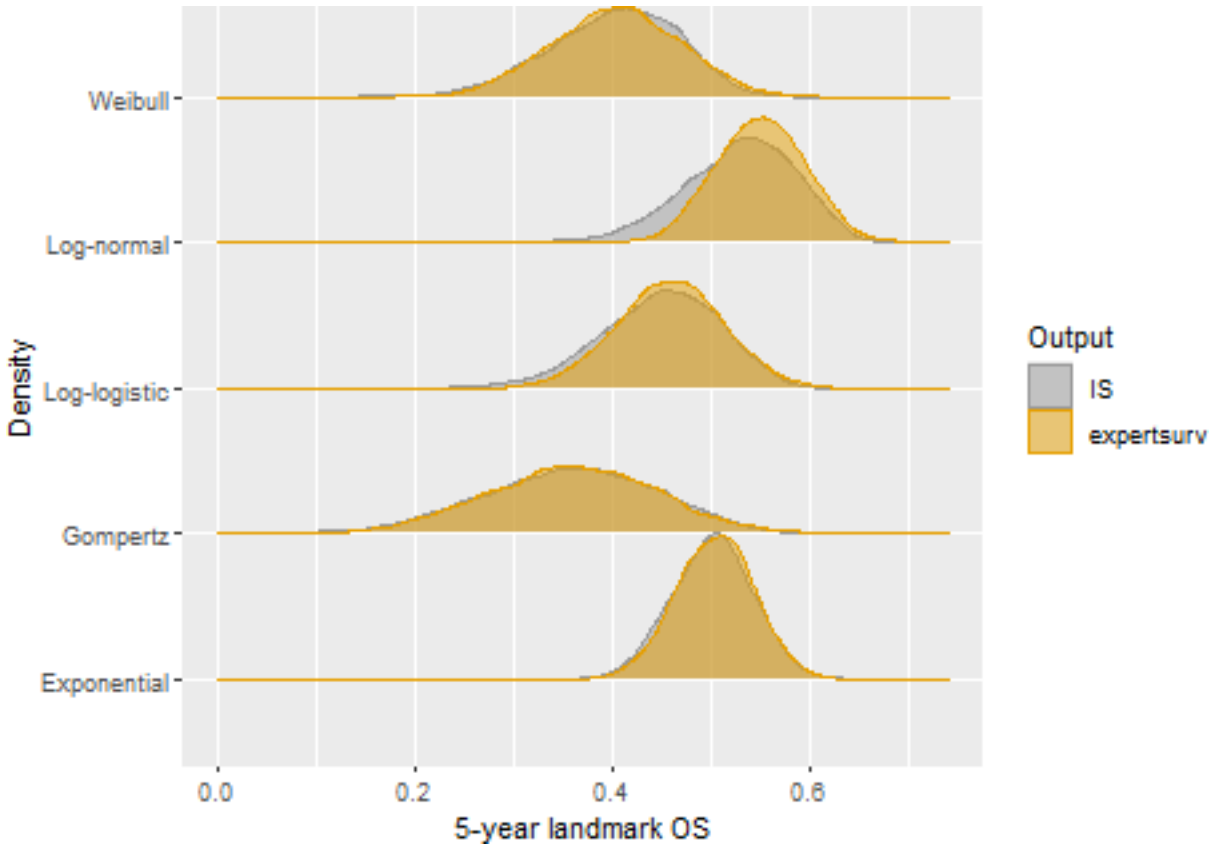


Figure 12: Density plots comparing 5-year OS estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with those obtained from the fully Bayesian method of Cooney and White implemented in the expertsurv package.